

Festo CMMT AS & CMMT ST Ring Topology MRP with Siemens PLC S7-1500 through Profinet

This Application Note explains how to implement Ring Topology in Festo CMMT-AS and CMMT-ST with Siemens S7-1500 series PLCs over a Profinet in the TIA programming environment.

It also describes how to incorporate LPNDR function block library into an TIA programming environment.

CMMT-ST-C8-1C-
MP-S0

CMMT-AS-C2-3A-
MP-S1

CMMT-AS-C4-3A-
MP-S1

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1 Introduction

This document explains how to connect multiple **CMMTs** in **ring topology** based on **MRP** with **Siemens S7- 1500 PLC** and then to get port diagnostics using function block **LPNDR_ReadGlobalInfo** from **LPNDR** library.

The application focuses on the following:

PLC (Siemens S7 1500) Configuration

- Configuration of CMMT AS & CMMT ST with Siemens S7-1500 PLC with MRP enabled in Ring Topology.
- Integration of **LPNDR** library into TIA v17 Project.
- Set up a PLC program.

1.1 Software Requirement

Software Name	Version (or) Firmware
TIA Portal	V17
CMMT-AS-C2-3A-MP-S1, CMMT-AS-C4-3A-MP-S1 and CMMT-ST-C8-1C-MP-S0 GSDML File	33.0.9.10

Table 1.1: Software Requirement



Note

- The “**LPNDR_ReadGlobalInfo**” library has been tested with TIA Portal v17.
- User should always ensure that they have the latest versions of software.

1.2 Hardware Requirement

Device Name	Part Number	Version (or) Firmware
Siemens S7-1500	6ES7 516-3FN01-0AB0	V2.9
Festo CMMT Servo Drive	CMMT-ST-C8-1C-MP-S0	V33.0.10.12
Festo CMMT Servo Drive	CMMT-AS-C2-3A-MP-S1	V33.0.10.12
Festo CMMT Servo Drive	CMMT-AS-C4-3A-MP-S1	V33.0.10.12
M12 – RJ45 Ethernet Cable	NEBC-D12G4-ES-1-S-R3G4-ET	Version Free

Table 1.2: Hardware Requirement

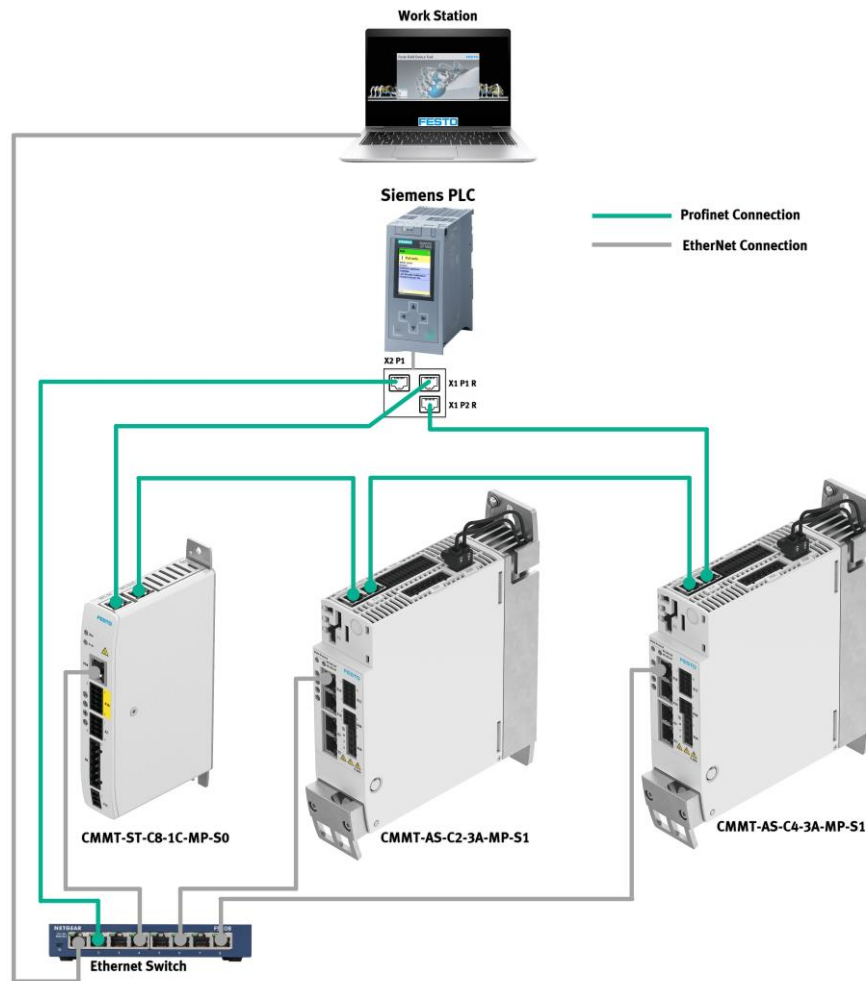


Note

- The hardware mentioned above are considered for this Application Note.

1.3 Topology Overview

Communication architecture of Siemens S7 1500 PLC and CMMT-ST and CMMT-AS using the Profinet protocol in a Ring Topology.



1.3.1 IP Table

Source Device	Port	IP Address / Subnet Mask
PC Work Station	Ethernet	192.168.0.50 / 255.255.255.0
Siemens S7 1500	X1	192.168.0.2 / 255.255.255.0
	X2	192.168.1.1 / 255.255.255.0
CMMT-ST-C8-1C-MP-S0	XF1	192.168.0.18 / 255.255.255.0
CMMT-AS-C2-3A-MP-S1	XF1	192.168.0.28 / 255.255.255.0
CMMT-AS-C4-3A-MP-S1	XF1	192.168.0.38 / 255.255.255.0

Table 1.3: IP Table



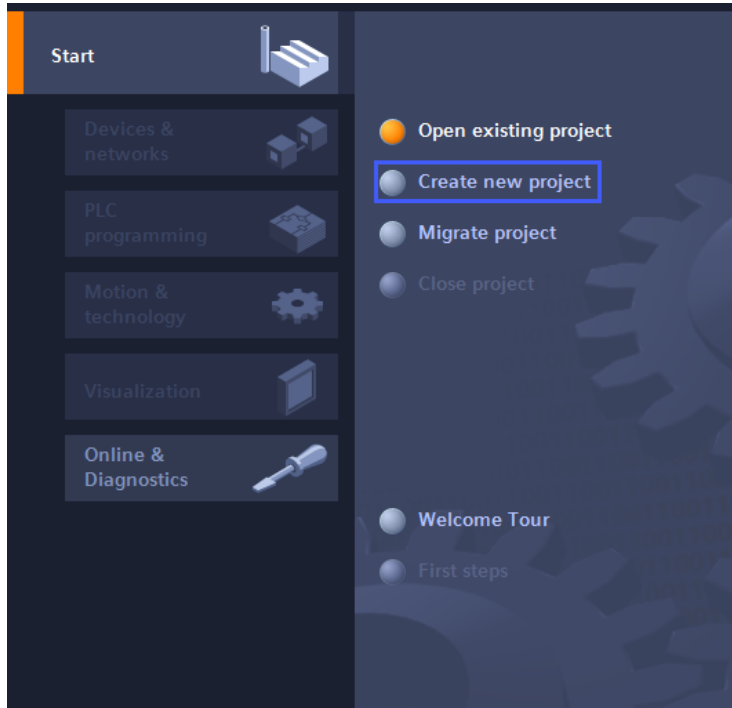
Note

- This application note is entirely based with respect to the above HW configuration as described.
- When setting up communication based on user network configuration, it's essential for the user to refer to the Festo documentation of CMMT ST and CMMT AS and appropriate Siemens documentation for the PLC.
- The above mentioned **IP addresses** and destination devices **are for this application note**. The user can assign IP addresses and destination devices as per their requirement.
- The ethernet Interface address of PLC , CMMTs in Ring Topology must be in the **same range**.

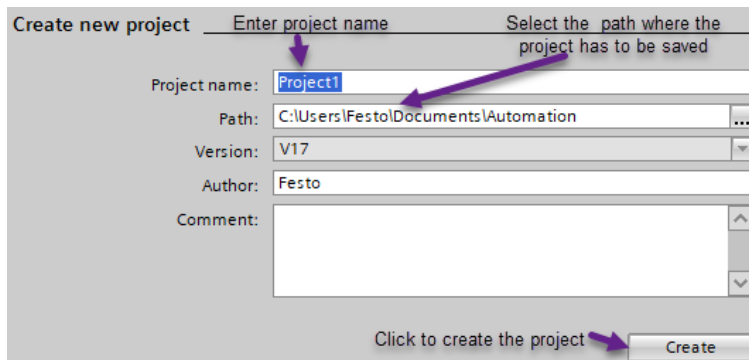
2 S7-1500 PLC Configuration in TIA Portal

2.1 Creating a new project in TIA Portal

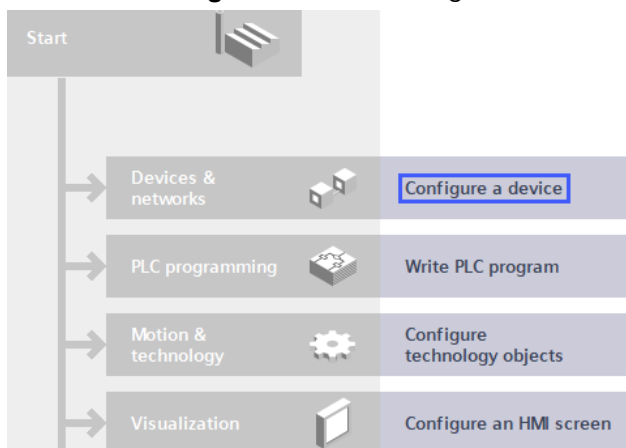
1. Start the **TIA Portal V17** software.
2. Click on **Create New Project** to create a new project.



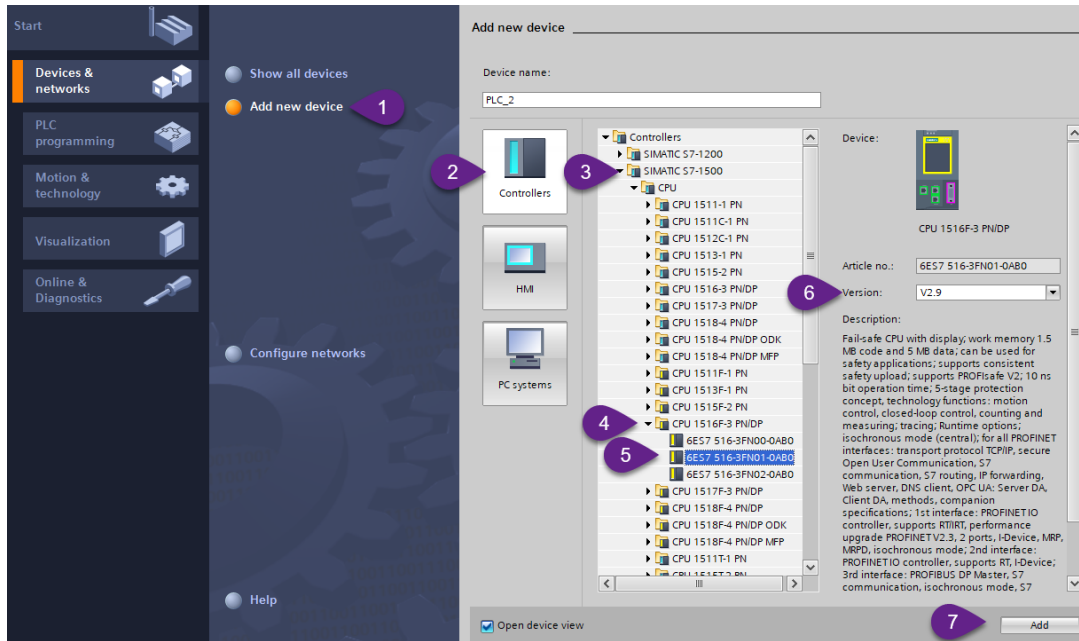
3. Enter the **Project name** and select the path where the project must be saved in the system. Click on **Create** to create the project.



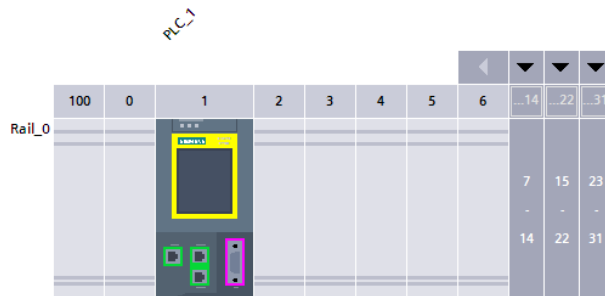
4. Click on **Configure a device** to configure the PLC needed.



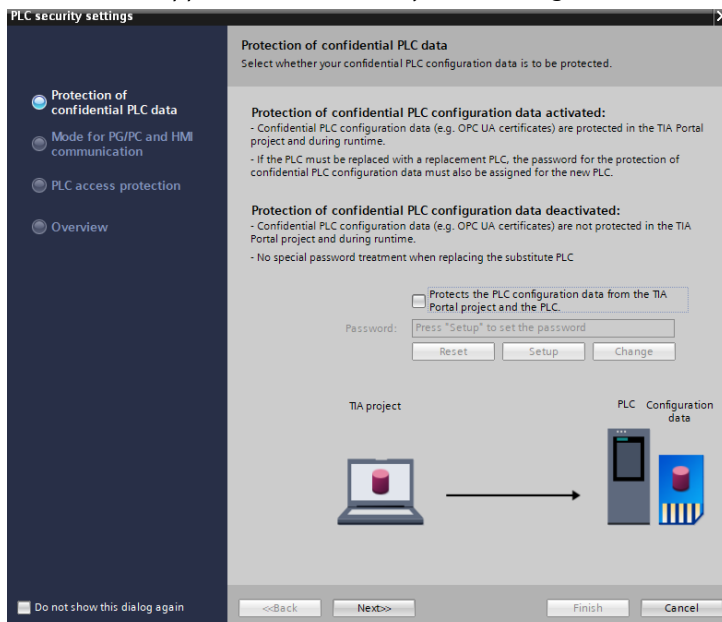
5. Steps for adding CPU to the project.



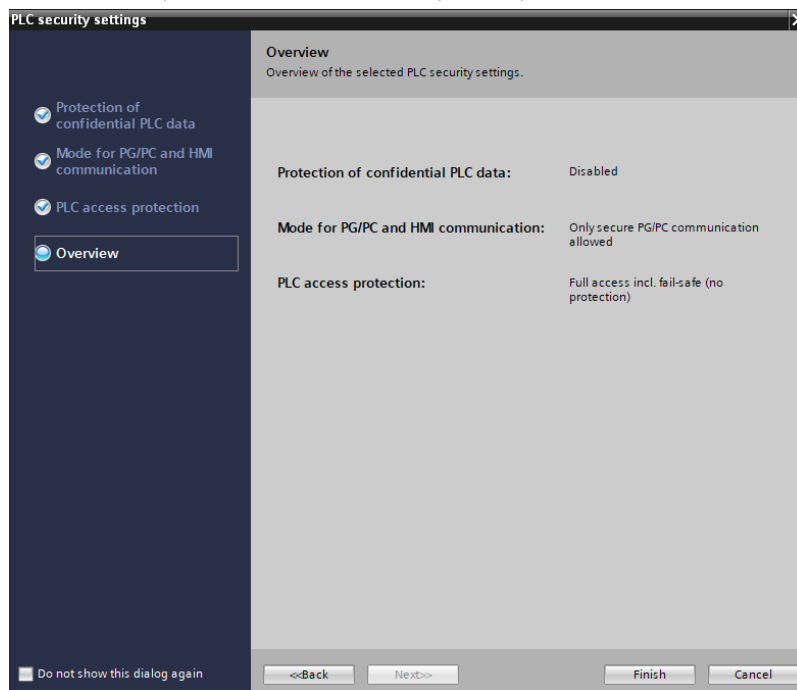
1. Click **Add new device**.
 2. Click **Controllers**.
 3. Expand **SIMATIC S7-1500**.
 4. Expand the CPU to view the CPUs.
 5. Select the required PLC from the list.
 6. Select the correct firmware version.
 7. Click **Add**.
6. **Project view** will be opened, PLC as selected will be shown. (Skip to [Section 2.2](#), if PLC is not a safety PLC)



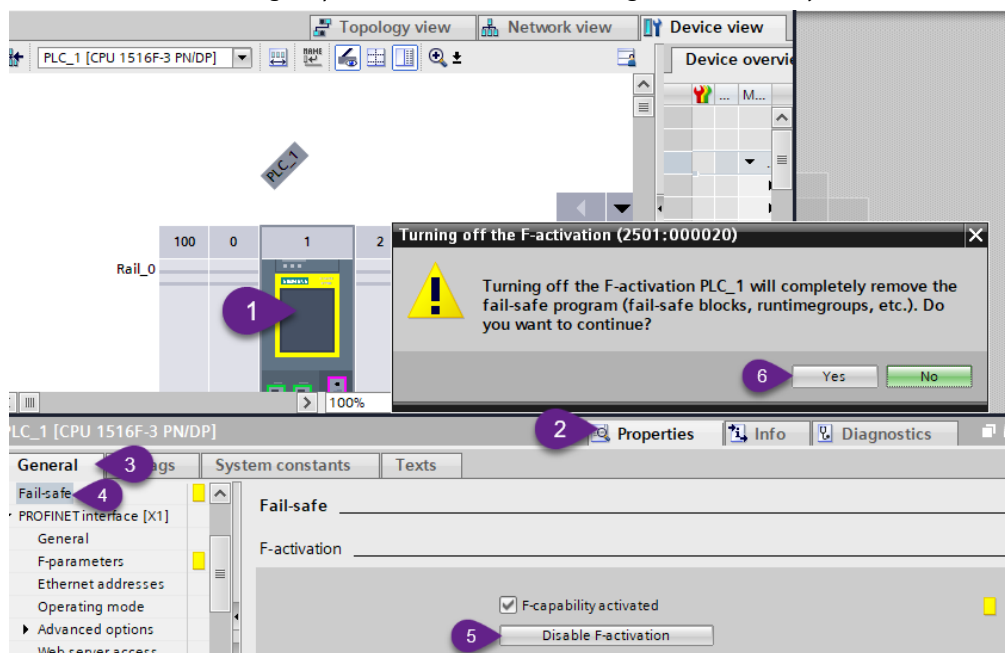
7. Since, in Application Note, safety PLC is being used. PLC security settings will be opened.



8. Here, safety has been disabled since, for the purpose of application note it is not needed. But, as per users' requirements it can be setup as required.



9. Follow the following steps to disable the **Fail-safe**, given that, safety has been disabled.



1. Double click, PLC CPU in **Device view**.
2. Click **Properties** tab.
3. Click **General** tab.
4. Click **Fail-safe** tab.
5. Click **Disable F-activation**.
6. Click **Yes**.

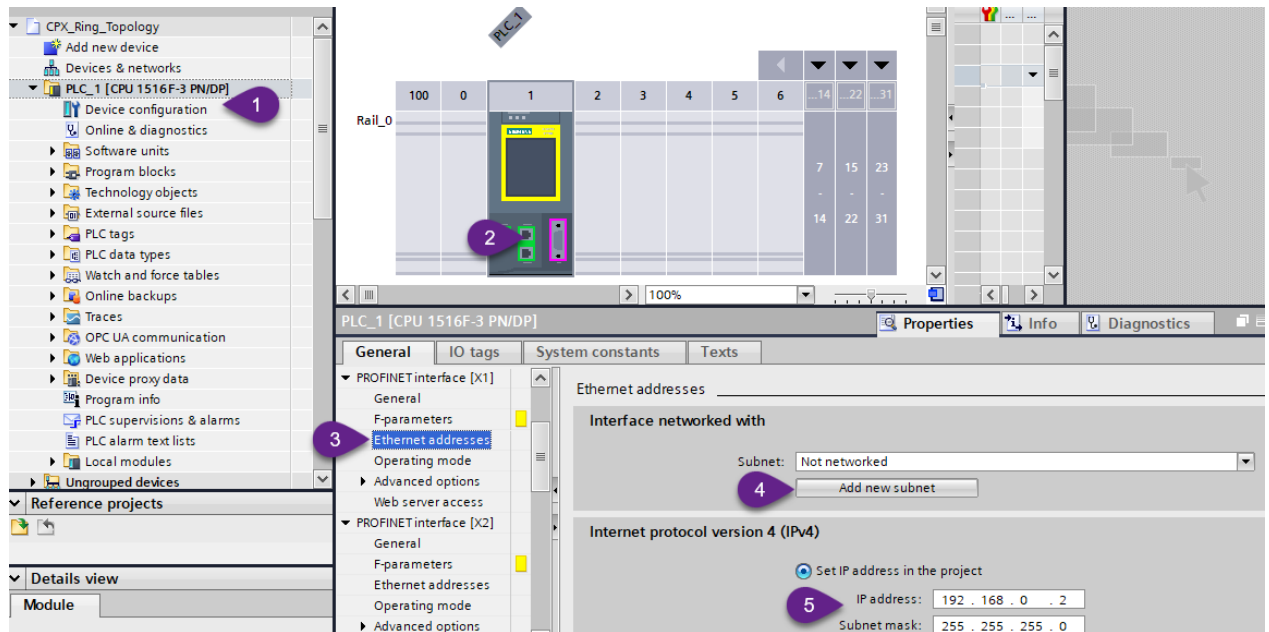


Note

- Users can use the method of detecting configuration of PLC directly without having to know the full model ID of the PLC CPU.

2.2 Configuration of the IP parameters of the Profinet interface of the PLC

- Follow the following steps to configure the IP parameters of the PLC.



1. Double click on **Device configuration**.
2. Double click on the Profinet interface to view **properties**.
3. Go to **General**, then click on **Ethernet addresses**.
4. Click **Add new Subnet**.
5. Enter the **IP address** and **subnet mask** as required.



Note

- Users has to make sure that Profinet addresses of devices in the ring and the PLC are in same range.

3 Configuration of CMMTs in TIA Portal

3.1 Adding the GSDML file to TIA Portal

3.1.1 Downloading the GSD file of CMMT drives from Festo Support Portal

1. Download the GSDML File for **CMMT** drives from the **Festo Support Portal**.



Note

- Users can search in search bar for getting GSDML files for modules according to requirements of the user.

Downloads for Servo drive

Product information 2 | User documentation 4 | Certificates 6 | **Software 14** | Expert knowledge 33 | Training 0

Filter in results

	Copy link	Download
Commissioning Festo Automation Suite Parameterisation, programming and maintenance of electronic devices by Festo	Link	Download
Function blocks CODESYS Modbus PTP function blocks for Codesys 3.5	Link	Download
Commissioning Festo Automation Suite - Plug-in Plug-in for the configuration and parametrisation of the servo drive CMMT-ST	Link	Download
Firmware Firmware Package Firmware and device description file for CMMT-ST-...-MP	Link	Download
Device Description Files PROFINET GSDML Supported systems: • servo drive CMMT-ST-C8-1C-MP-S0 (8163946) More		

File Name	Version	SHA-512	Size	Download
GSDML.zip	V32.0.10.10	SHA-512	18 KB	Download
GSDML-CMMT-ST-MP.zip	33.0.9.10	SHA-512	40 KB	Download

2. In the webpage, click **Downloads** tab, then **Software** tab, then open **Device Description Files** and Download the latest version.
3. Save the GSDML file in an appropriate folder in the PC, it will be downloaded as a zipped file as shown in the image below.



GSDML-CMMT-AS-MP.zip

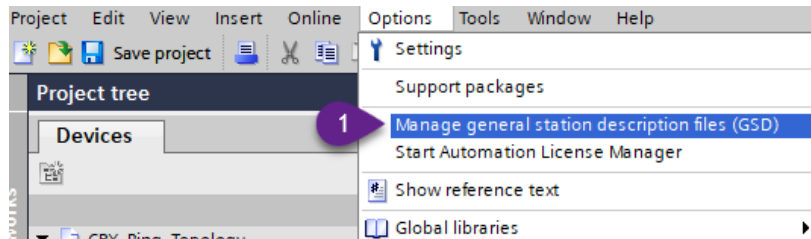


GSDML-CMMT-ST-MP.zip

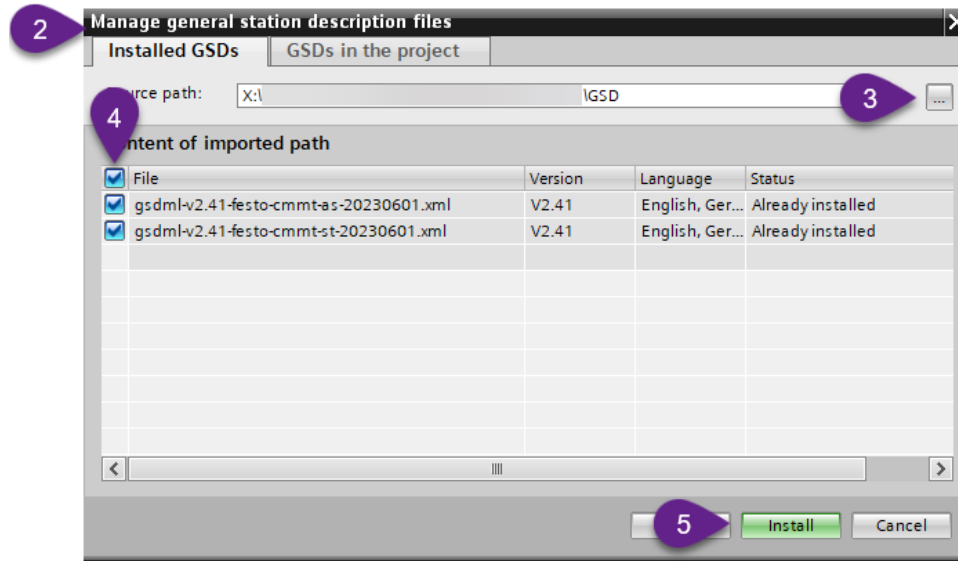
4. Unzip the files in order to add them into TIA Portal.

3.1.2 Adding the download GSDML file to TIA Portal

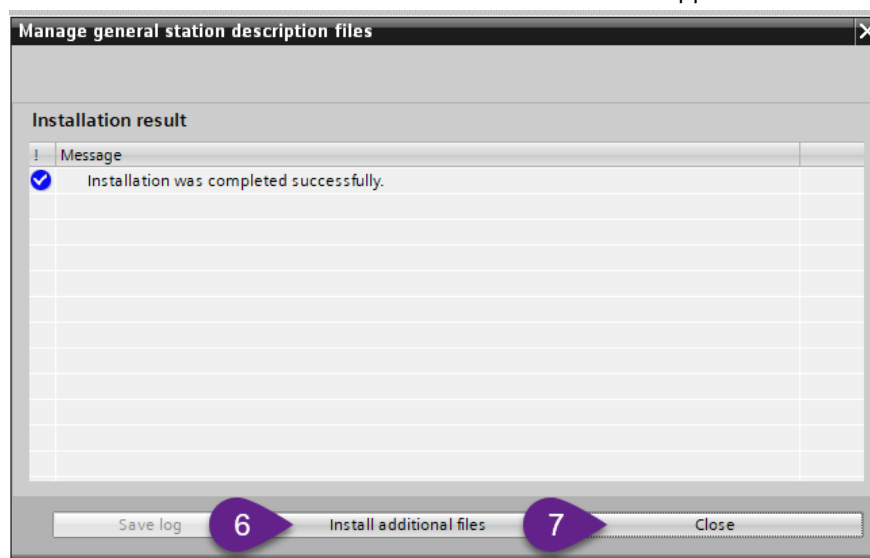
1. Click **Options** then **Manage General Station Description Files(GSD)**.



2. Window will pop-up as shown in the image below.



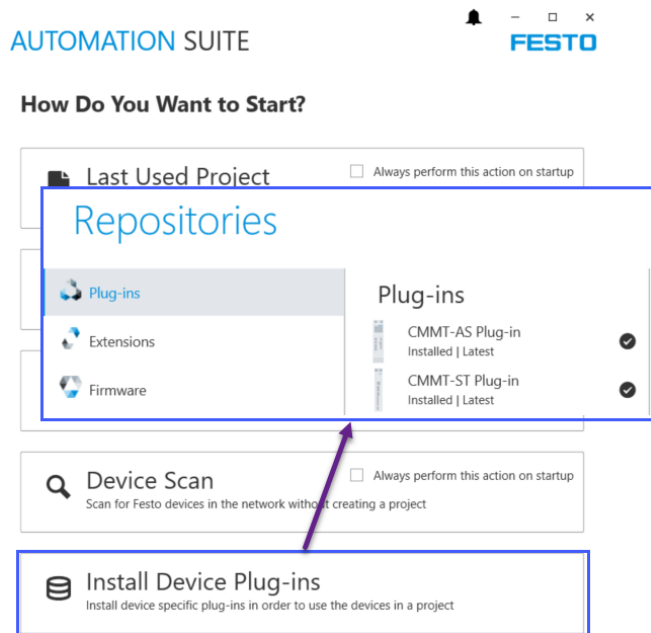
3. In **Source path**, browse to where GSDML files were unzipped, then select the folder and click **OK**.
4. Select the File as shown in the image, make sure it is showing correct module ID.
5. Click **Install**.
6. Once that is done, click **Install additional files**, then repeat the steps 2 to 4 in order to add in GSDML files for other drives which has been download and unzipped.



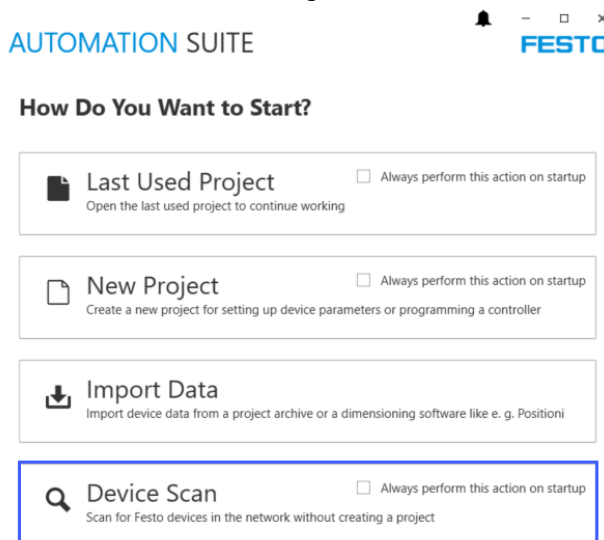
7. Click **Close**, once required GSDML files have been added.

3.2 Configuring CMMT drives in Festo Automation Suite

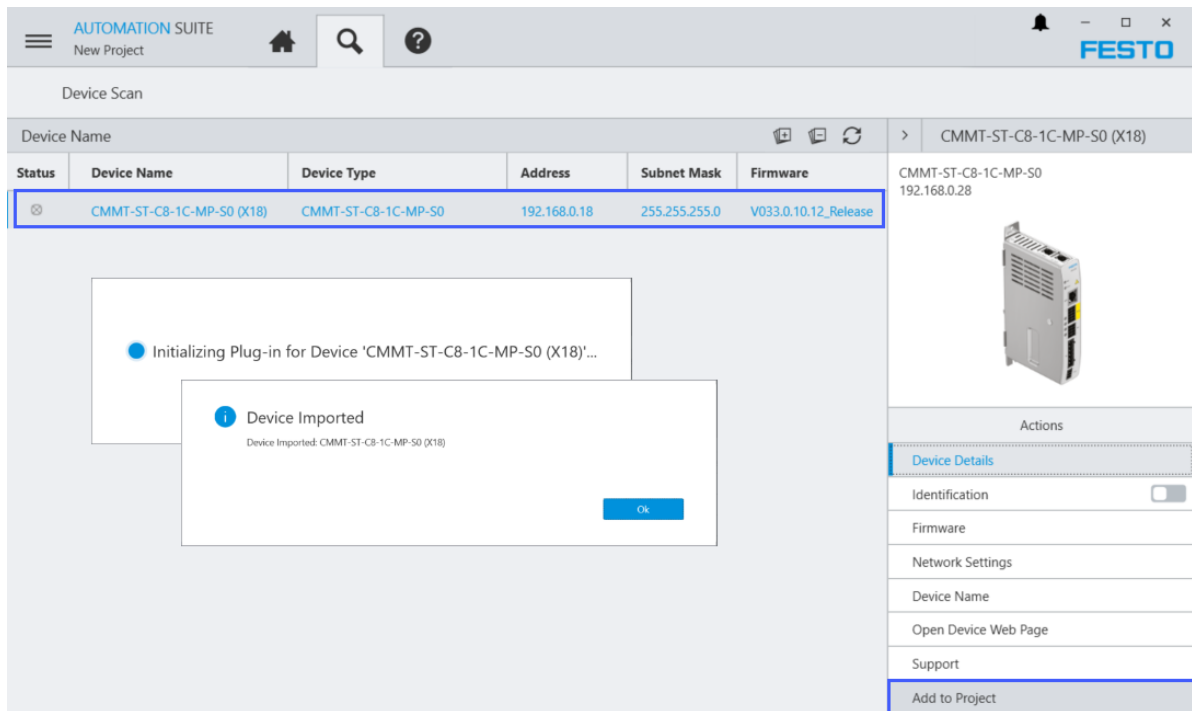
- Since, the CMMT drives being used are multi-protocol ones, it is possible that they may have different communication protocol enabled other than Profinet one. So, it is recommended to check if CMMTs are using Profinet communication protocol first through the use of Festo Automation suite.
 - Do the following steps in order to check if Profinet protocol is enabled, if not change it to Profinet.
1. Make sure to check if X18 port of CMMT to be configured is connected to PC Workstation having Festo Automation Suite.
 2. Make sure to have Plug-ins for the CMMT are installed in the Festo Automation Suite.



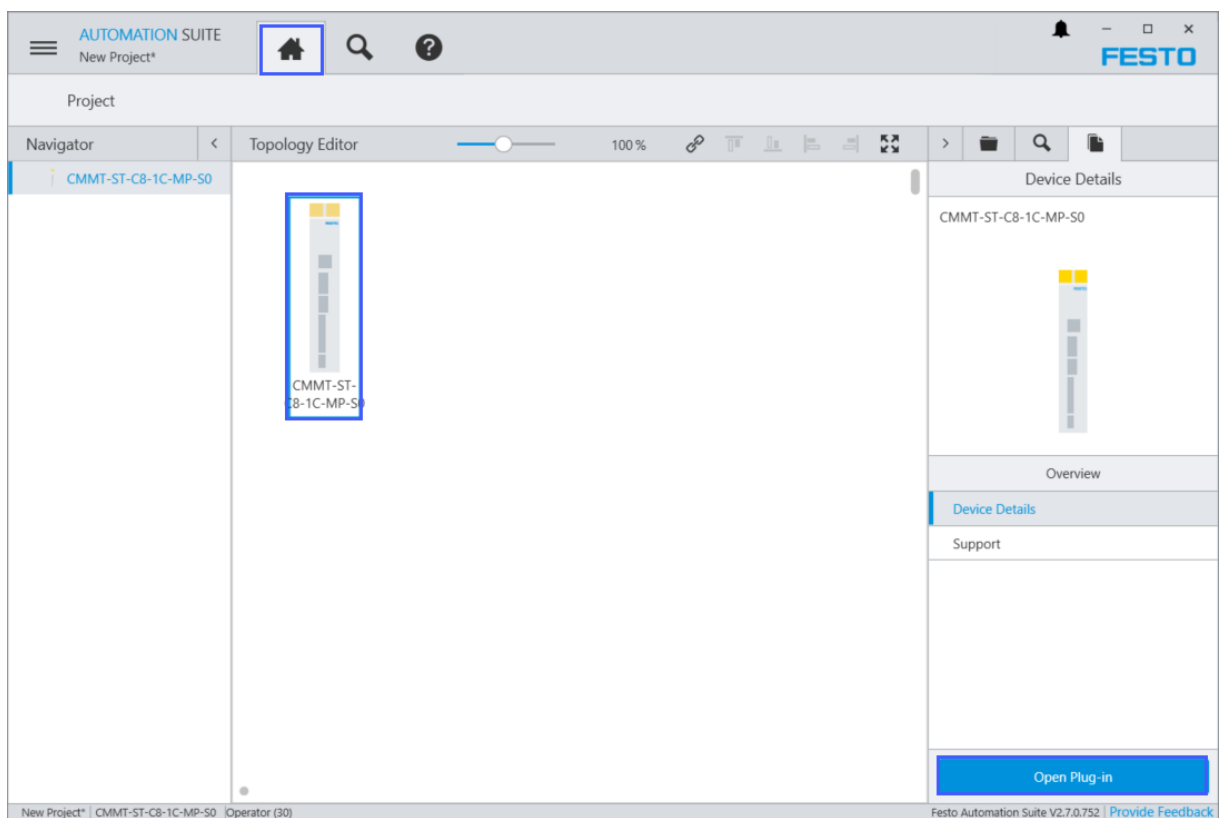
3. Click Device Scan, if Plug-in is installed.



- Click the CMMT(here, CMMT-ST), then Click Add to Project, then Click OK after Device gets Imported.



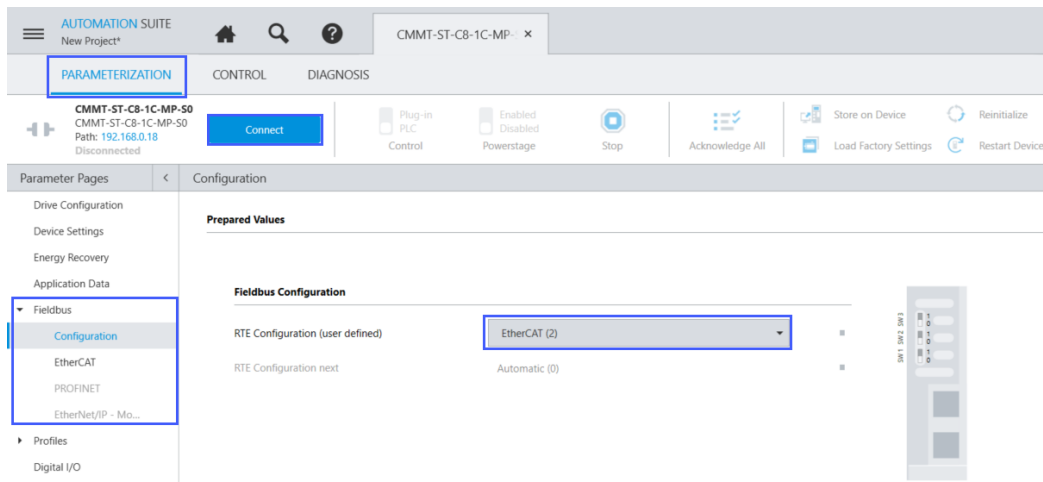
- Click Home Button, then either double click the Device in the Topology Editor or click once then click Open Plug-in.



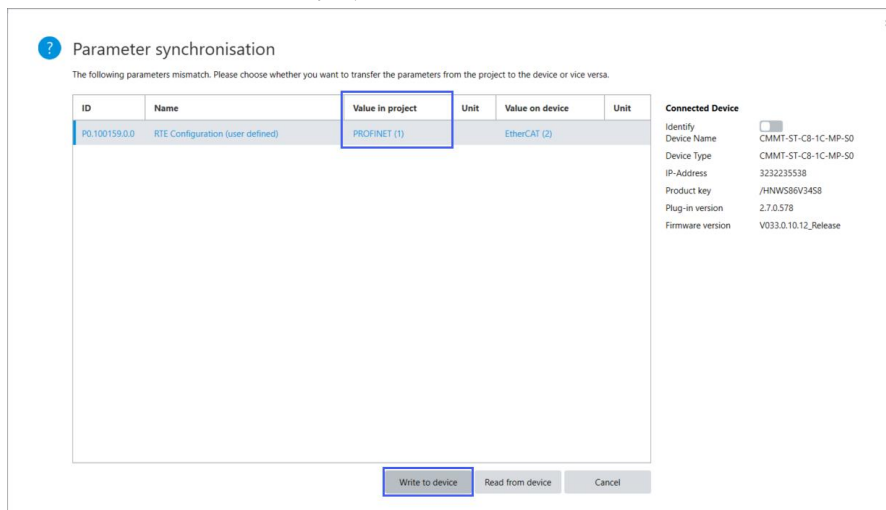
- Screen will open as shown in the image below. In Parameterization tab, Click IP address in blue, then select the CMMT drive, then Click Apply.



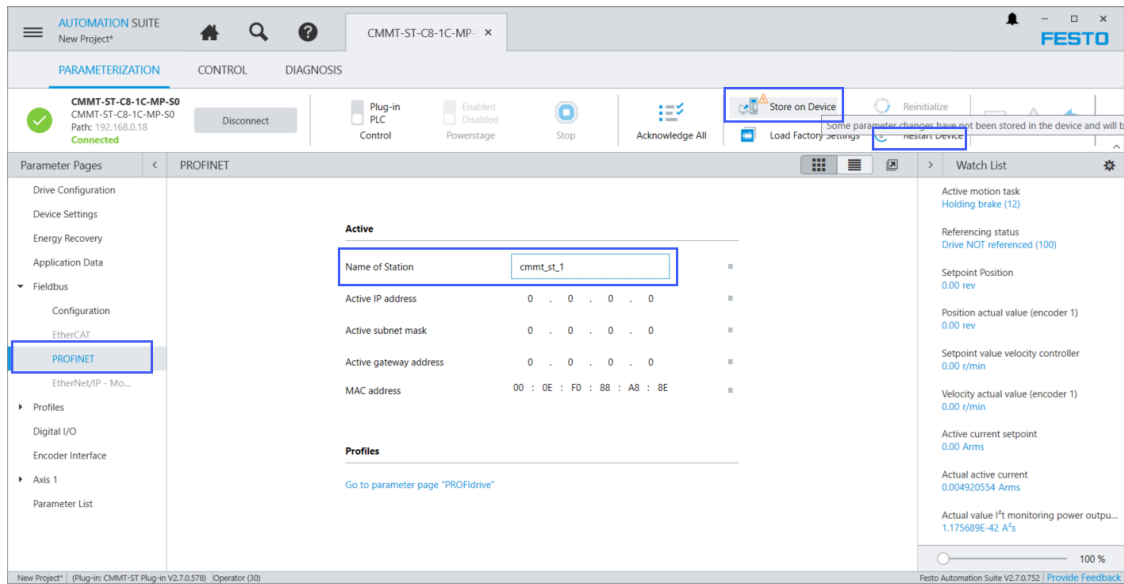
- Expand Fieldbus, then Click Configuration, then in RTE Configuration (user defined), select Profinet (1) and Click connect.



- Make sure the Value in project is Profinet (1), then click Write to device.



9. Once, the CMMT is connected, in Profinet Section inside Fieldbus, Name of the Station can be set according to the user requirement, then Store on Device can be clicked and after that Restart Device to finalise the configuration.



10. CMMT is now ready to be used for the further configuration into ring in TIA portal through use of Profinet.
11. All other CMMTs can be configured in same way, follow the same steps as shown in [Section 3.2](#).



Note

- Make sure to connect the CMMT drives from their X18 port to PC work station to configure them in Festo Automation Suite.

3.3 Configuring the CMMT drives in TIA Portal

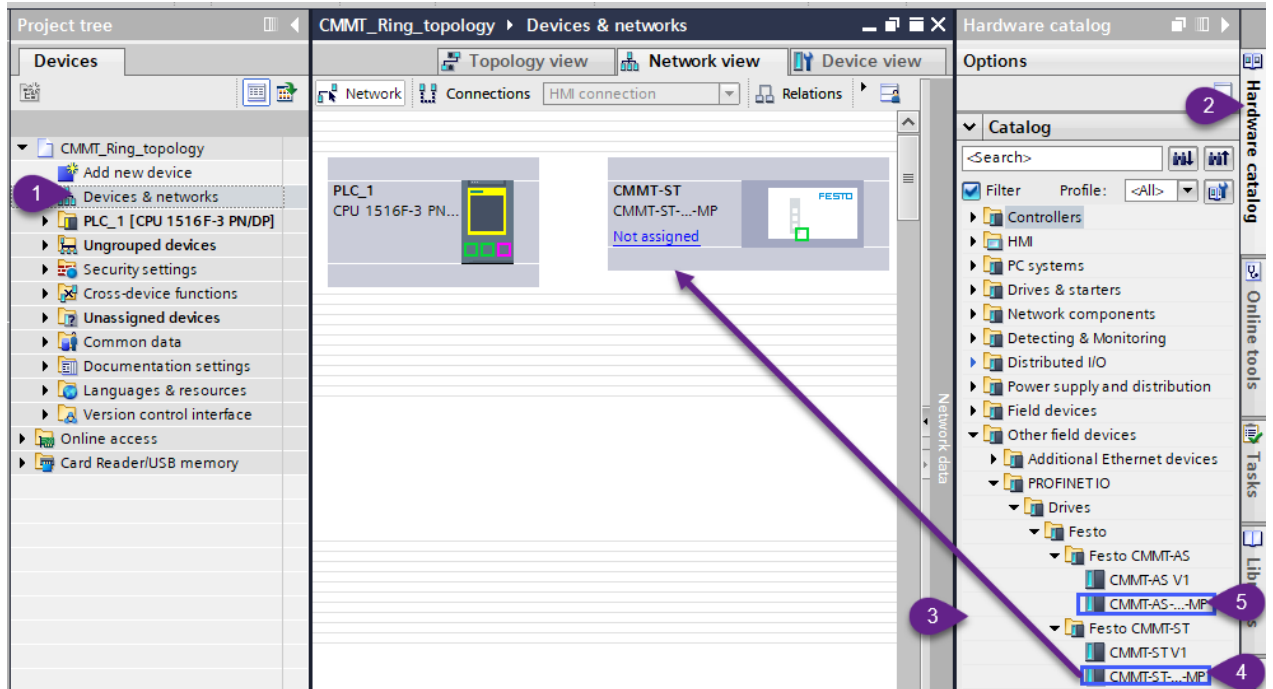


Note

- Before, connecting the drives to Ring, make sure to configure each drive one by one to PLC.
- When configuration of drives are being done, make sure that Work Station is connected to X1, and one side of ring(shouldn't be closed yet) to other Port of X1 of Siemens S7-1500.
- Only close the ring, after all the drives have been configured. At this time, Work station should be connected to X2.

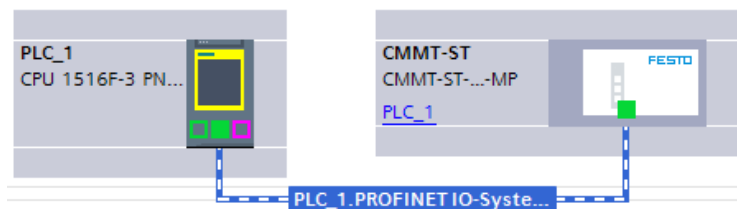
3.3.1 Adding the installed modules to Network View

- Do the following steps to add installed modules to the Project.



1. Double click on **Devices and Networks**, then open **Network view**.
2. Click **Hardware catalog**.
3. Locate the GSDML of CMMT drives under **Other Field Devices >>> Profinet IO >>> Drives >>> Festo >>> Festo CMMT-ST >>> CMMT-ST-...-MP**.
4. Drag and Drop the module as shown in the image to **Network view**.
5. Do the same for CMMT-AS drives as shown in the image.

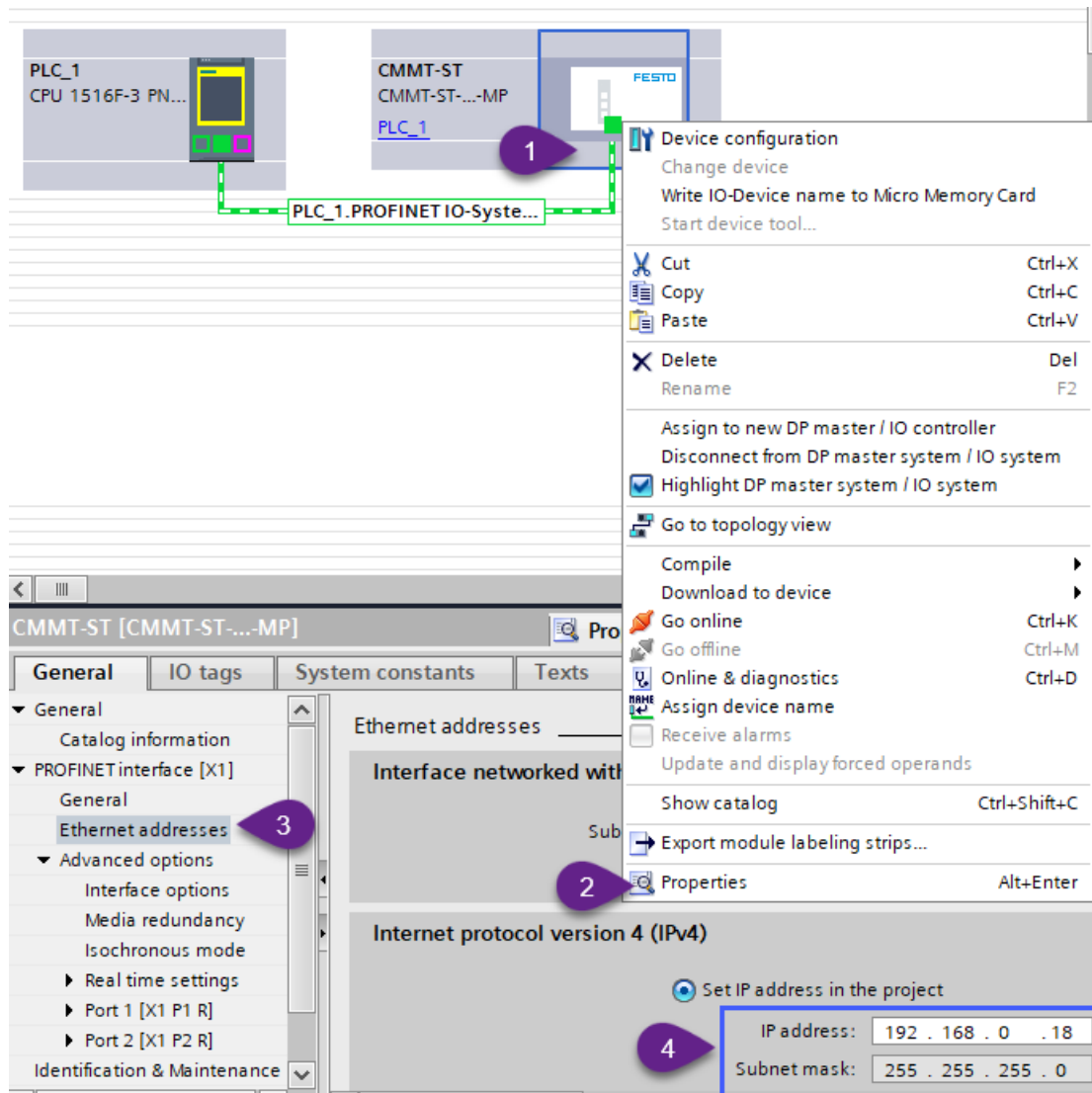
- Connect the S7-1500 PLC to CMMT-ST Drive. In the example, CMMT-ST drive is connected to PROFINET interface[X1] of S7-1500 PLC.



3.3.2 Network Configuration of Profinet interface CMMT drives in TIA Portal

In TIA portal,

- Do the following steps to do network configuration of Profinet interface of the module.



1. Right click on **CMMT-ST**.
2. Click on **Properties**.
3. Go to **Ethernet addresses**.
4. Set the **IP address** as per user's requirements.

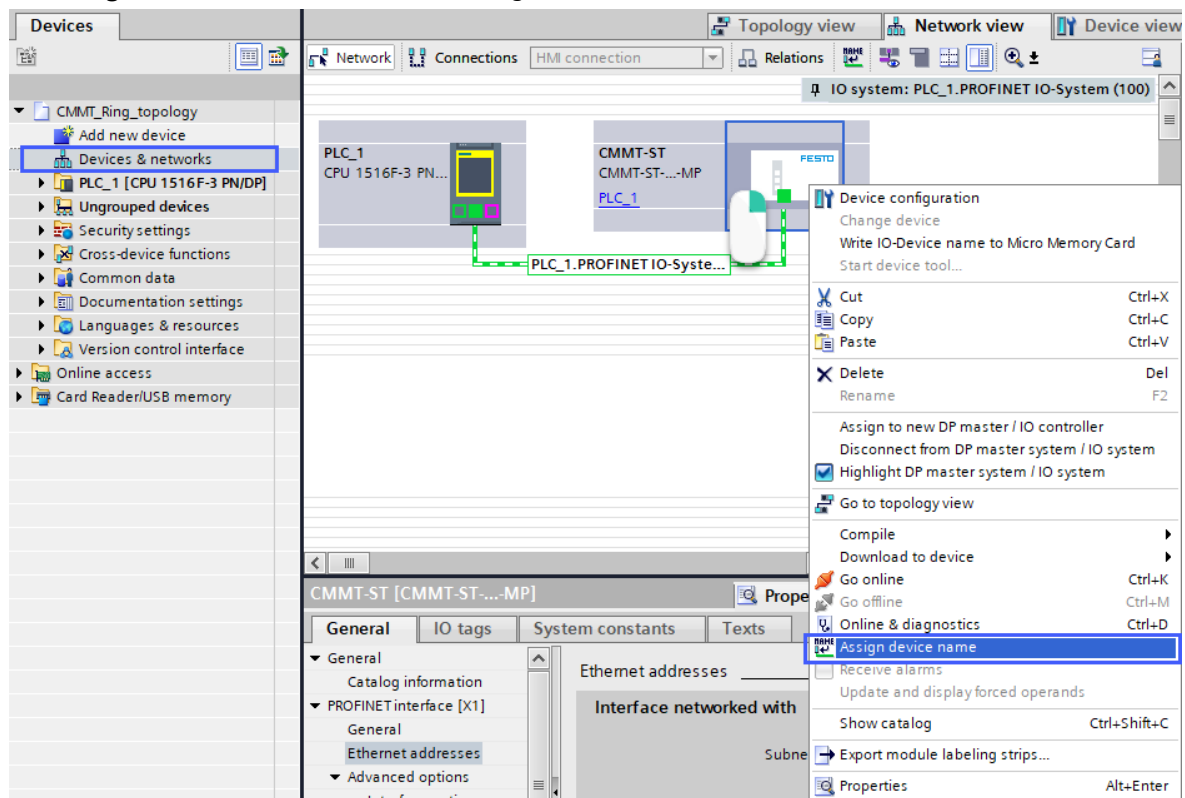


Note

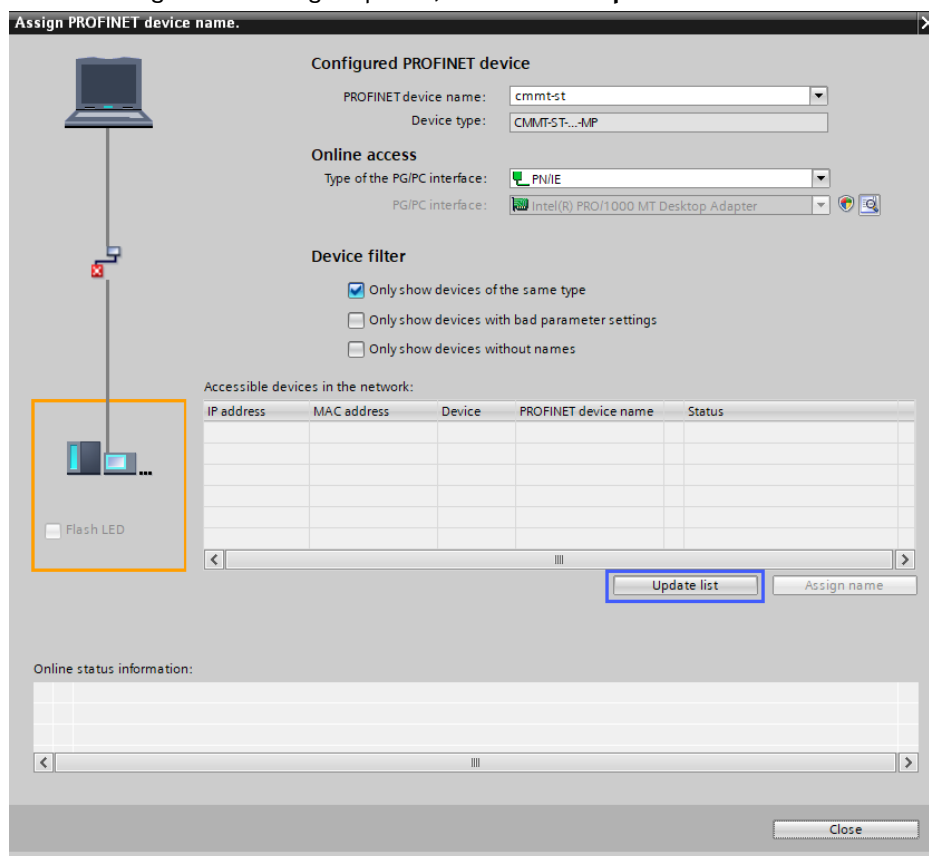
- The IP address of the Profinet interface of the modules, along with PLC must be in the same range.

3.3.3 Assigning Profinet name to CMMT Drives

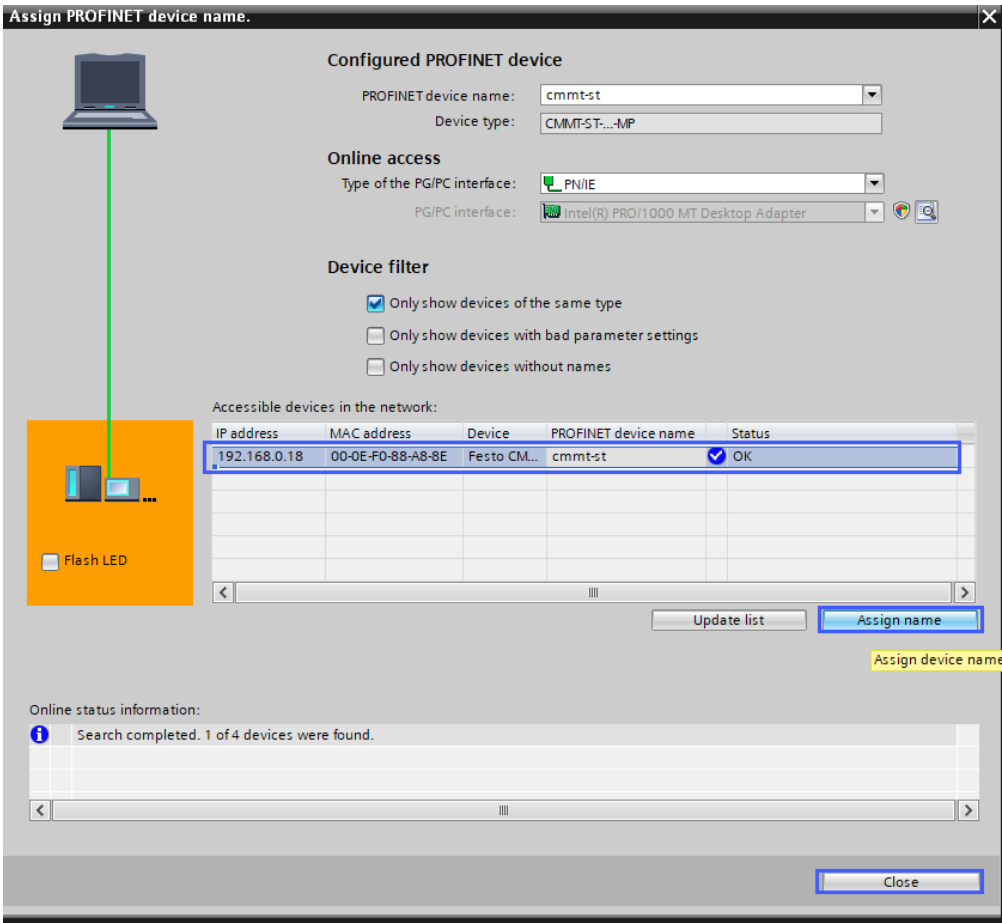
1. Assign device name by going to **Devices & networks**, then right clicking **CMMT-ST** icon, then clicking **Assign device name** as shown in the image below.



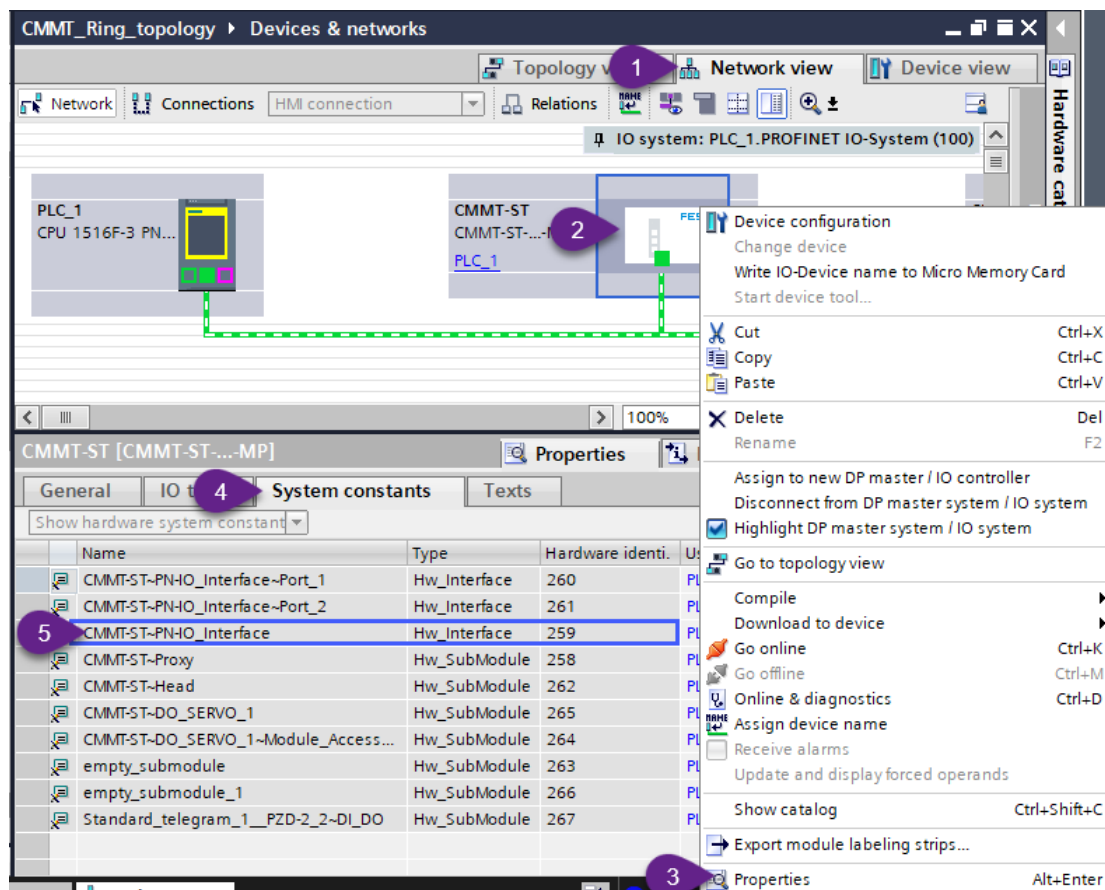
2. Following window will get opened, then click on **Update list**.



3. Once the list is updated, the device(s) as shown below will appear, then click on it before clicking **Assign name**. Once that is done, Status should be shown as **OK**. Then, **Close** can be clicked.



3.3.4 Determining Hardware ID of Devices



1. Open **Network view**.
2. Right click the drive for which **Hardware ID** is to be found.
3. Click on **Properties**.
4. Click **System constants** tab.
5. Here, user can note down the Hardware ID of the device as required, likewise, user can repeat all the steps for each drive accordingly.



Note

- Since, for the purpose of this Application note, diagnostics of Port is being looked at, Hardware ID of PN-IO Interface is being used.
- For PLC, Local Profinet Interface 1 is being monitored since those Ports are the ones which are in the Ring.
- Follow the [Section 3.2](#), for configuring any other drives which user wants to add in the ring. (Here, for the purpose of Application Note, two CMMT-AS also has been added to the ring).

4 Ring(MRPD) Topology Setup and Testing

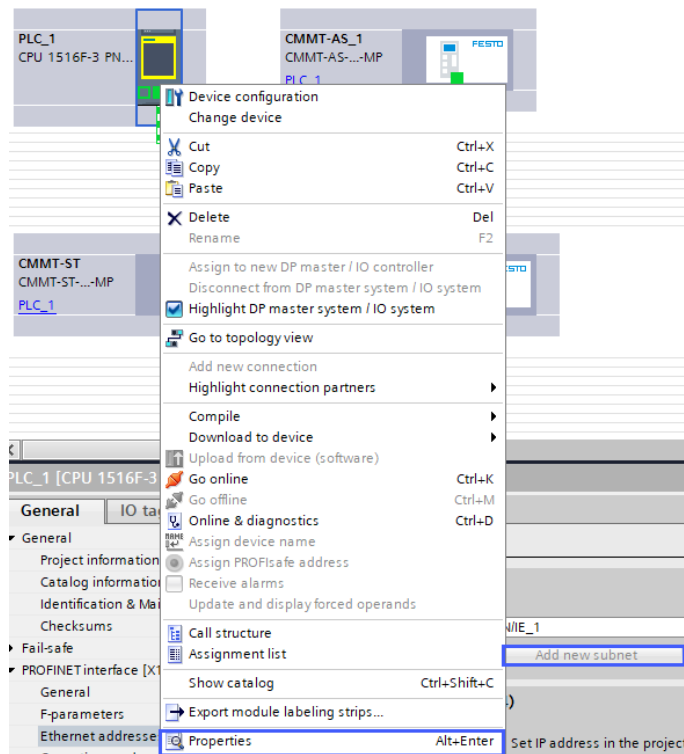
4.1 Ring Topology Setup



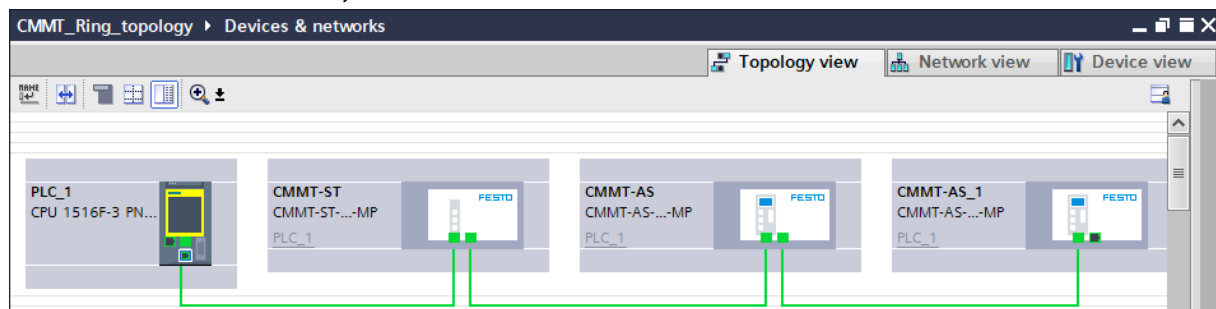
Note

- Users can go through Setup of Ring Topology based on "MRP" in Siemens Portal in order to get full understanding of how Ring with redundancy is setup with Siemens PLCs.
- Users should now connect Work Station to X2, with X1(P1 R and P2 R) for the Ring.

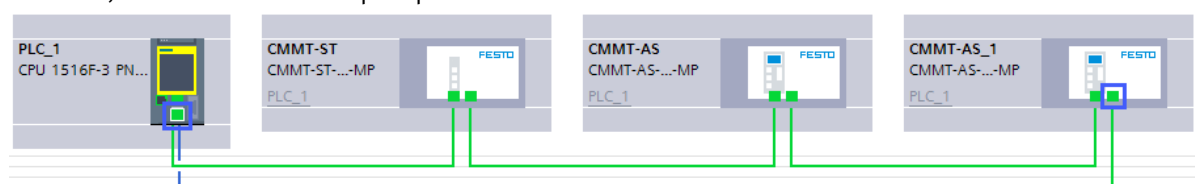
1. First thing should be to make sure that, **new subnet** has been added for PLCs and other devices in the ring.



2. In **Topology view** in **Devices & networks**, connect devices and module in a line to correct ports first based on how it is actually connected.



3. Now, connect the non-occupied ports of PLC and the final module.



4. Now, configure the ring's **MRP domains** by going to **Network view**, then clicking **PN/IE_1** (Make sure **Highlight IO system** is unchecked), then **Properties** then **MRP domains** >>> **Devices** then Set PLC as the **Manager** and all others as **Client**.

The screenshot shows the 'PN/IE_1 [Industrial Ethernet]' properties window. The 'MRP domains' section is expanded, showing 'mrpdomain-1' and its 'Devices' table. The table lists the devices and their roles in the MRP domain.

	PROFINET device name	MRP domain	MRP role	Ring port 1	Ring port 2
1	cmmt-as	mrpdomain-1	Client	Port 1 [X1 P1 R]	Port 2 [X1 P...
2	cmmt-as_1	mrpdomain-1	Client	Port 1 [X1 P1 R]	Port 2 [X1 P2 R]
3	cmmt-st	mrpdomain-1	Client	Port 1 [X1 P1 R]	Port 2 [X1 P2 R]
4	plc_1.profinet interface_1	mrpdomain-1	Manager	Port_1 [X1 P1 R]	Port_2 [X1 P2 R]

5. Next, go to **Sync domains** >>> **Devices** then in **IO devices**, Make sure **RT class** is **RT** for all.

The screenshot shows the 'PN/IE_1 [Industrial Ethernet]' properties window. The 'Sync domains' section is expanded, showing 'Sync-Domain_1' and its 'IO devices' table. The table lists the devices and their RT class.

	PROFINET device name	RT class	Synchronization role	Redundancy level
	plc_1.profinet interface_1	RT...	Unsynchronized	
	cmmt-as	RT	Unsynchronized	
	cmmt-as_1	RT	Unsynchronized	
	cmmt-st	RT	Unsynchronized	

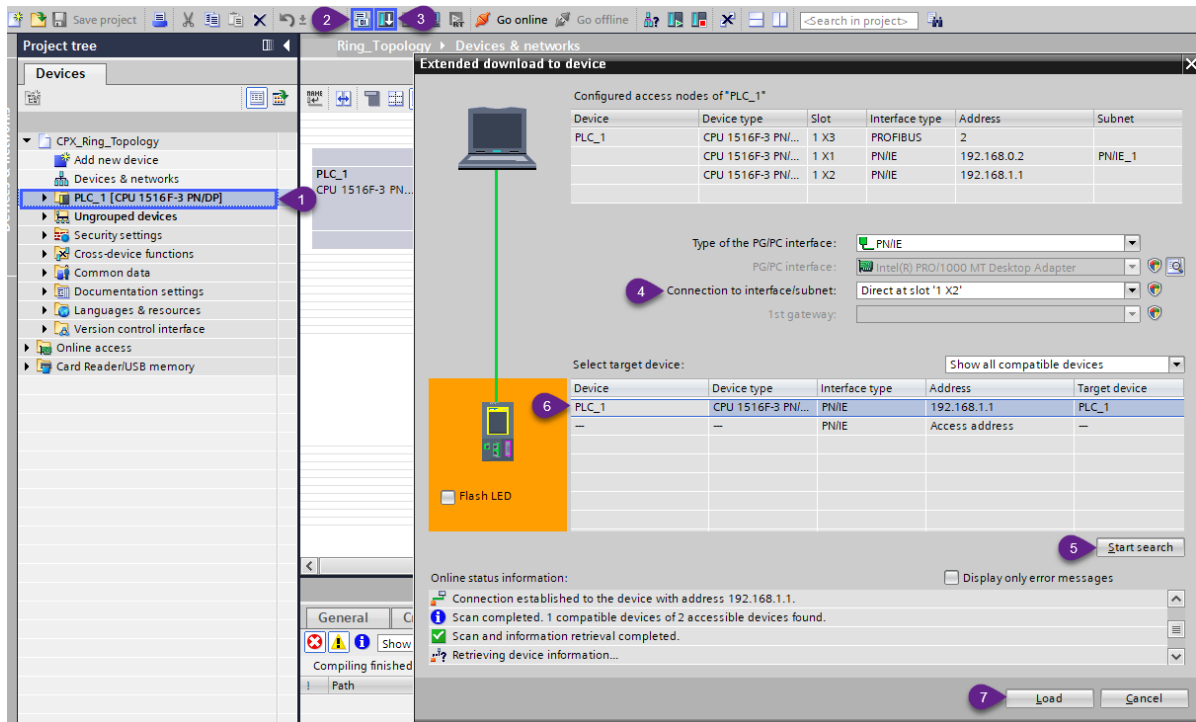


Note

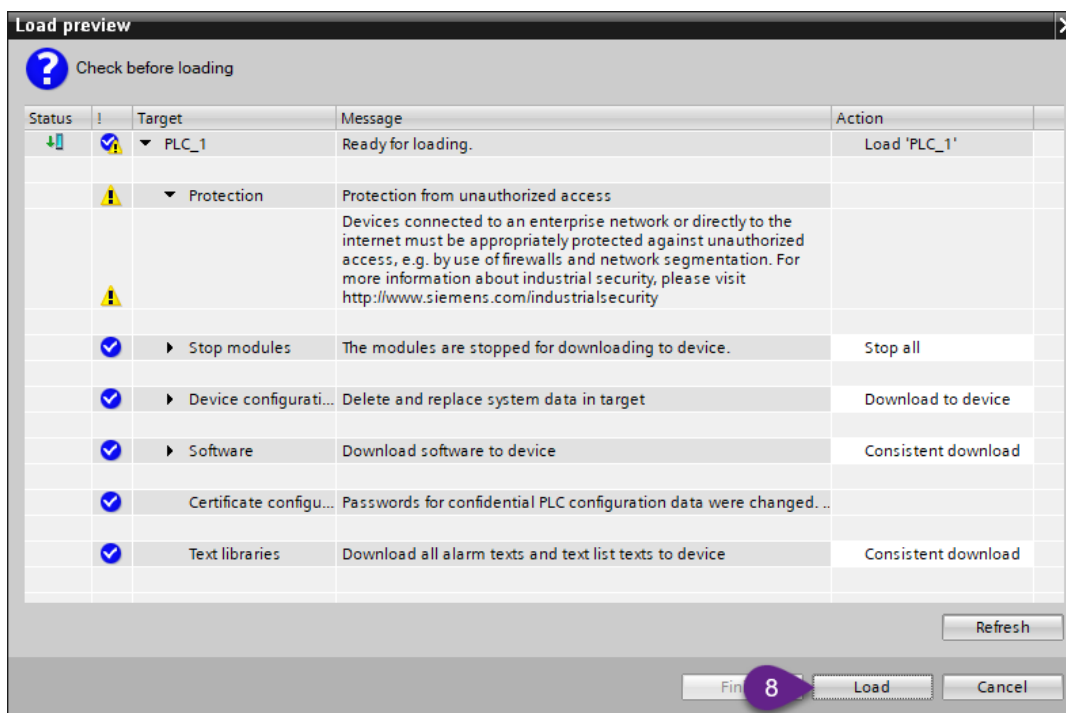
- Users can go with **RT class** as **IRT**, given that **MRP** is set as **Not in a Ring** configuration, and Ring Topology is not used.

4.2 Downloading to PLC and Testing

- Do the following steps to Compile and Downloading the configuration into the PLC.



1. Click on **PLC**.
2. **Compile** can be clicked now. There should be no errors after compiling, given that, all the previous steps have been followed without any problem.
3. Once that is done, user can **Download** the configuration into PLC and Click on **Download**.
4. Select the slot from which Work Station and PLC are linked(Here it is X2, since both the slots of X1 are forming a Ring with IO modules).
5. Click **Start search**.
6. Here, PLC with X2's IP address will be visible(there will be no errors here, given that connections and IP configurations are done properly).
7. Click **Load**.

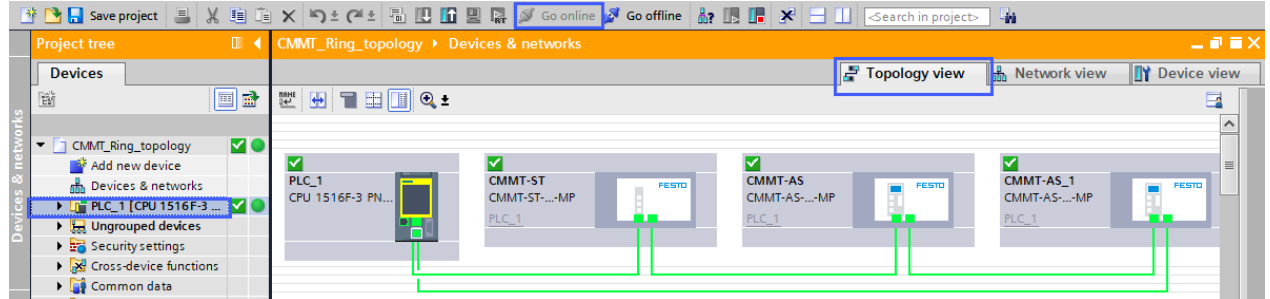


- There will be **Load preview** as show in the image above. Click **Load** here as well.

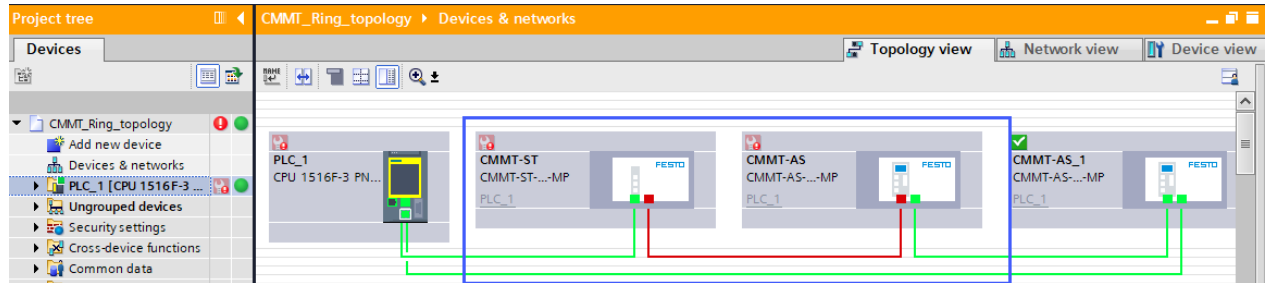
Thus, the downloading will be completed.

- Now, the basic testing can be done.

- First User has to **Go online**, here it should be all green with all connections in the **Topology View**.



- Then, try removing hardware connections to see if feedback is there.



- Connect them back, it should be resolved automatically.

5 Adding LPNDR Library to the Project

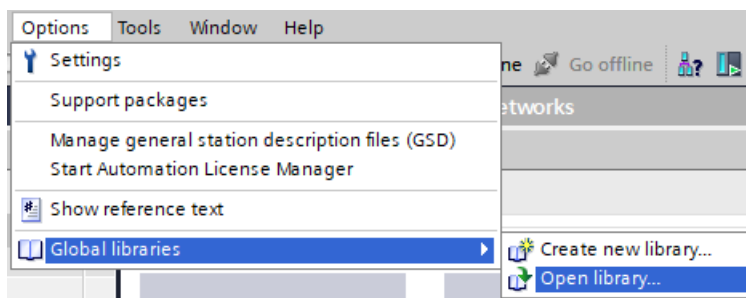
- LPNDR Library is a Library for Profinet data records. It has function blocks block which are used to read carious device and status information from a Profinet device and describe the parameters. For, the purpose of this application note, **ReadGlobalInfo** function block of LPNDR Library has been used in order to get diagnostic details of Profinet connections and ports in the Ring Network.



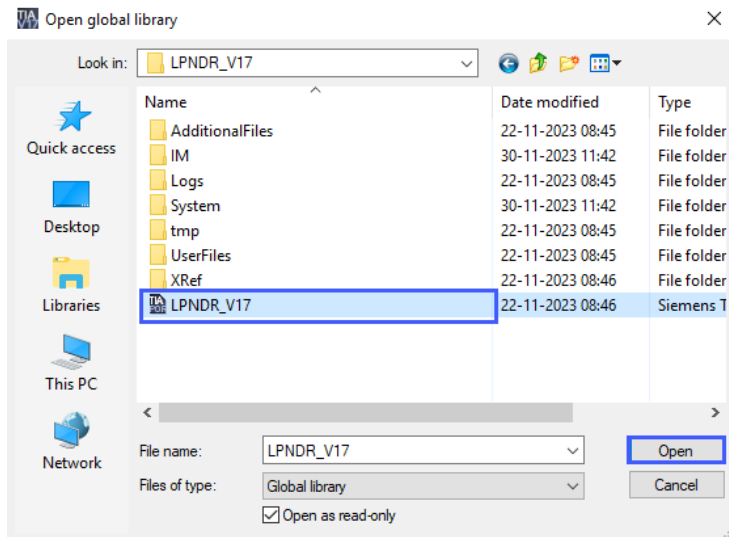
Note

- **LPNDR library** for S7-1500 can be downloaded from the Siemens support.
- User can go through LPNDR documentation in the Siemens support for details of blocks in the Library.

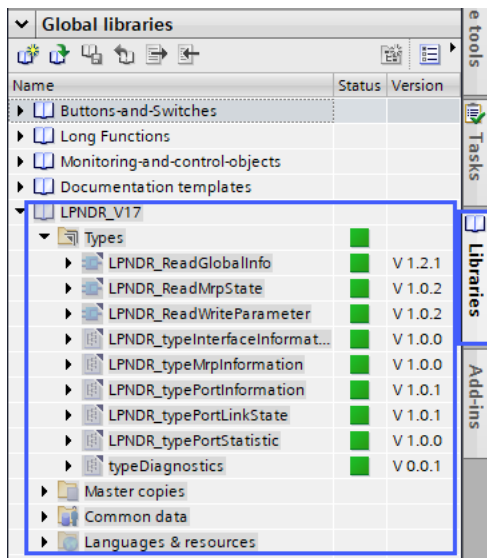
1. Unzip the library downloaded from the support portal into a folder of user's choice.
2. Click on **Option > Global libraries > Open library**.



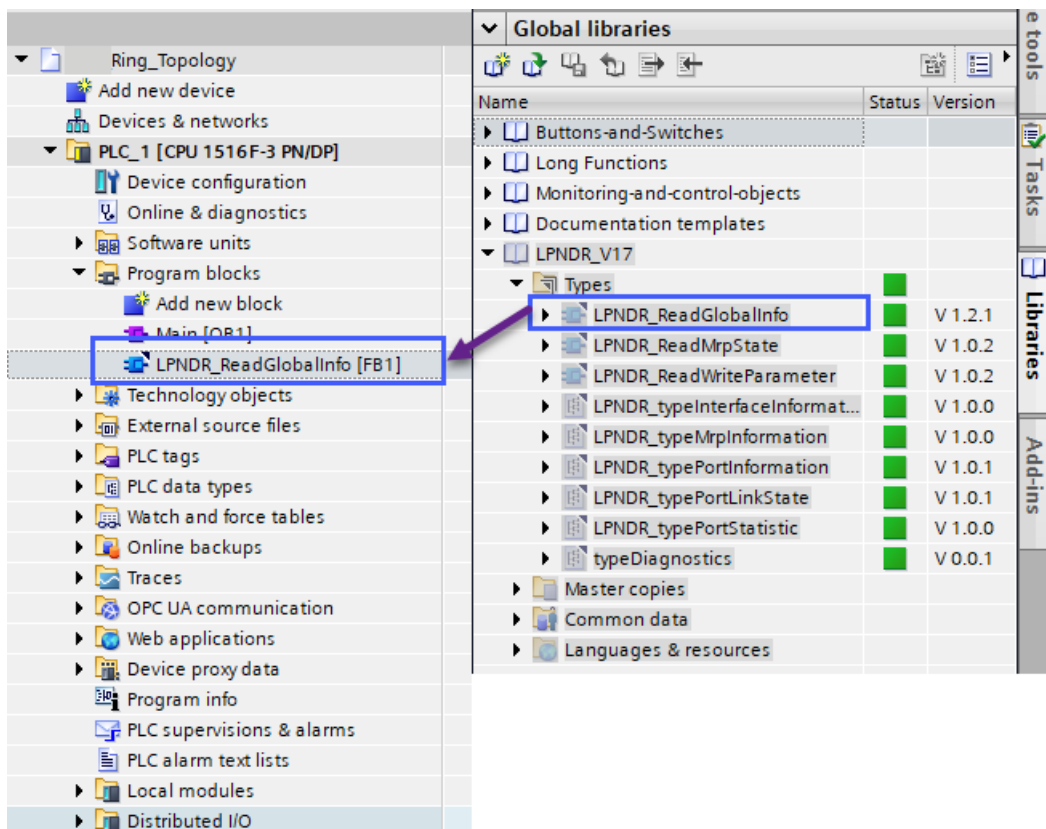
3. Navigate through the path where library has been unzipped in the system. Then, select the library and click **Open**.



4. In TIA Portal, open **Libraries** and navigate to **LPNDR** and open to view its contents.



5. Drag and drop **LPNDR_ReadGlobalInfo** from **Libraries** to **Program blocks** section the project.



6. Based on user requirements, other blocks can be implemented as well. As for this application note purpose, only this one block is implemented.



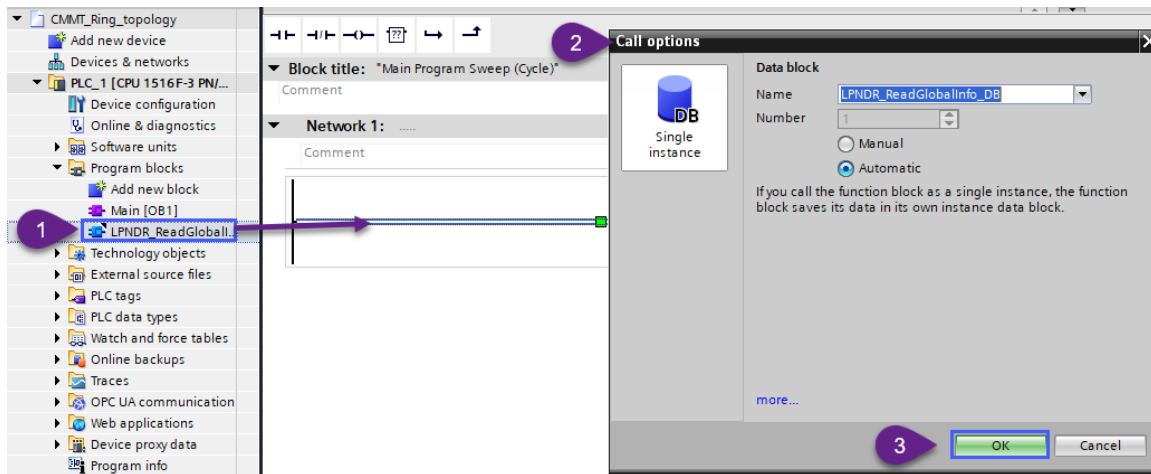
Note

- User can add more of Function blocks, as per the requirements, for Application note purpose, 3 sets of function blocks have been used to get Port diagnostics of 3 CMMT drives connected in the ring.

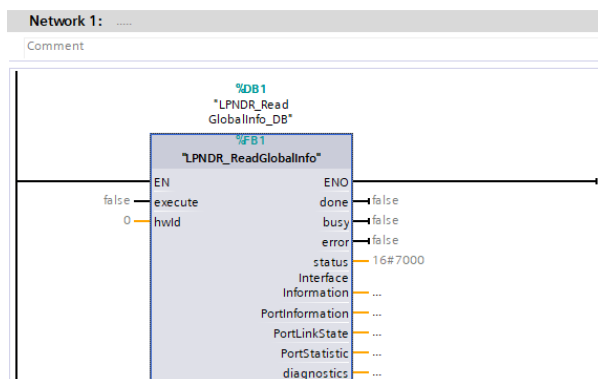
6 Configuration of LPNDR_ReadGlobalInfo function block in TIA Project

6.1 Configuring the Input and Output Tags

1. Drag and drop function block into **Network** inside **Main [OB1]**.



2. **Call options** will be opened, enter the name of FB instance.
3. Click **OK** to confirm.
4. The called Function Block will be as shown below:

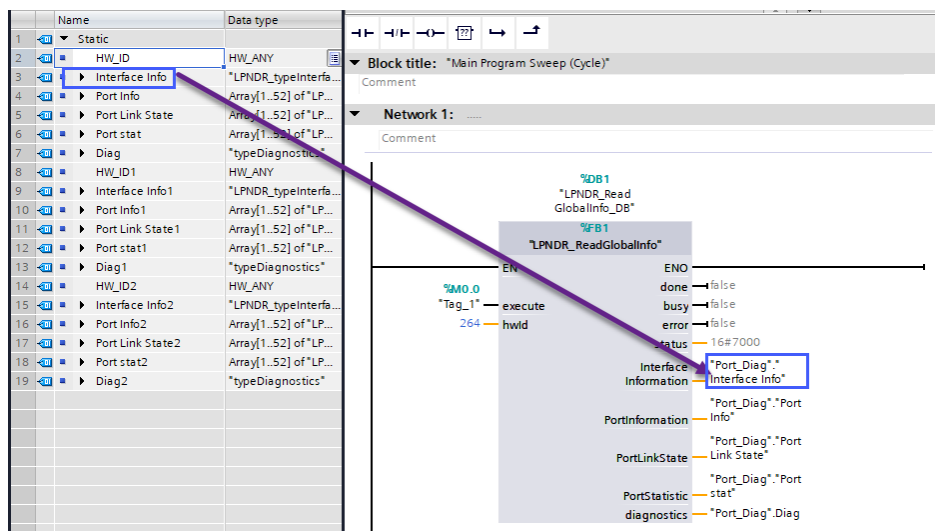


5. Create a **Data block** for assigning tags in the Function block according to data types from function block.

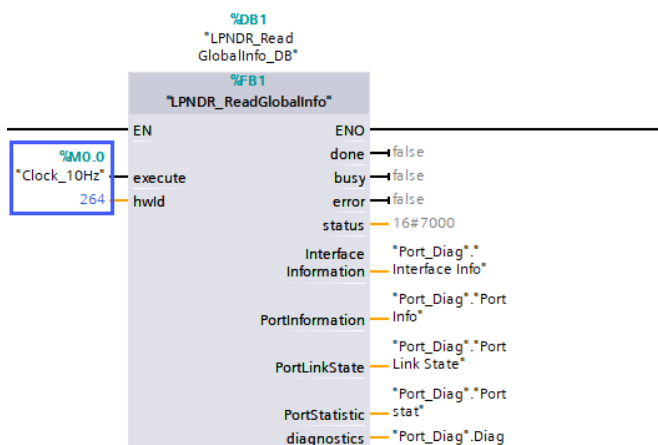
Project tree: CMMT_Ring_topology > PLC_1 [CPU 1516F-3 PN/DP] > Program blocks > Port_Diag [DB4]

Name	Data type	Start value	Retain	Accessible f...	Write...	Visible in ...	Setpoint
1 Static							
2 HW_ID	HW_ANY	0					
3 Interface Info	"LPNDR_typeInterfa...						
4 Port Info	Array[1..52] of "LP...						
5 Port Link State	Array[1..52] of "LP...						
6 Port stat	Array[1..52] of "LP...						
7 Diag	"typeDiagnostics"						
8 HW_ID1	HW_ANY	0					
9 Interface Info1	"LPNDR_typeInterfa...						
10 Port Info1	Array[1..52] of "LP...						
11 Port Link State1	Array[1..52] of "LP...						
12 Port stat1	Array[1..52] of "LP...						
13 Diag1	"typeDiagnostics"						
14 HW_ID2	HW_ANY	0					
15 Interface Info2	"LPNDR_typeInterfa...						
16 Port Info2	Array[1..52] of "LP...						
17 Port Link State2	Array[1..52] of "LP...						
18 Port stat2	Array[1..52] of "LP...						
19 Diag2	"typeDiagnostics"						
20 HW_ID3	HW_ANY	0					
21 Interface Info3	"LPNDR_typeInterfa...						
22 Port Info3	Array[1..52] of "LP...						
23 Port Link State3	Array[1..52] of "LP...						
24 Port stat3	Array[1..52] of "LP...						
25 Diag3	"typeDiagnostics"						

- Drag and drop them to respective places in the function block.

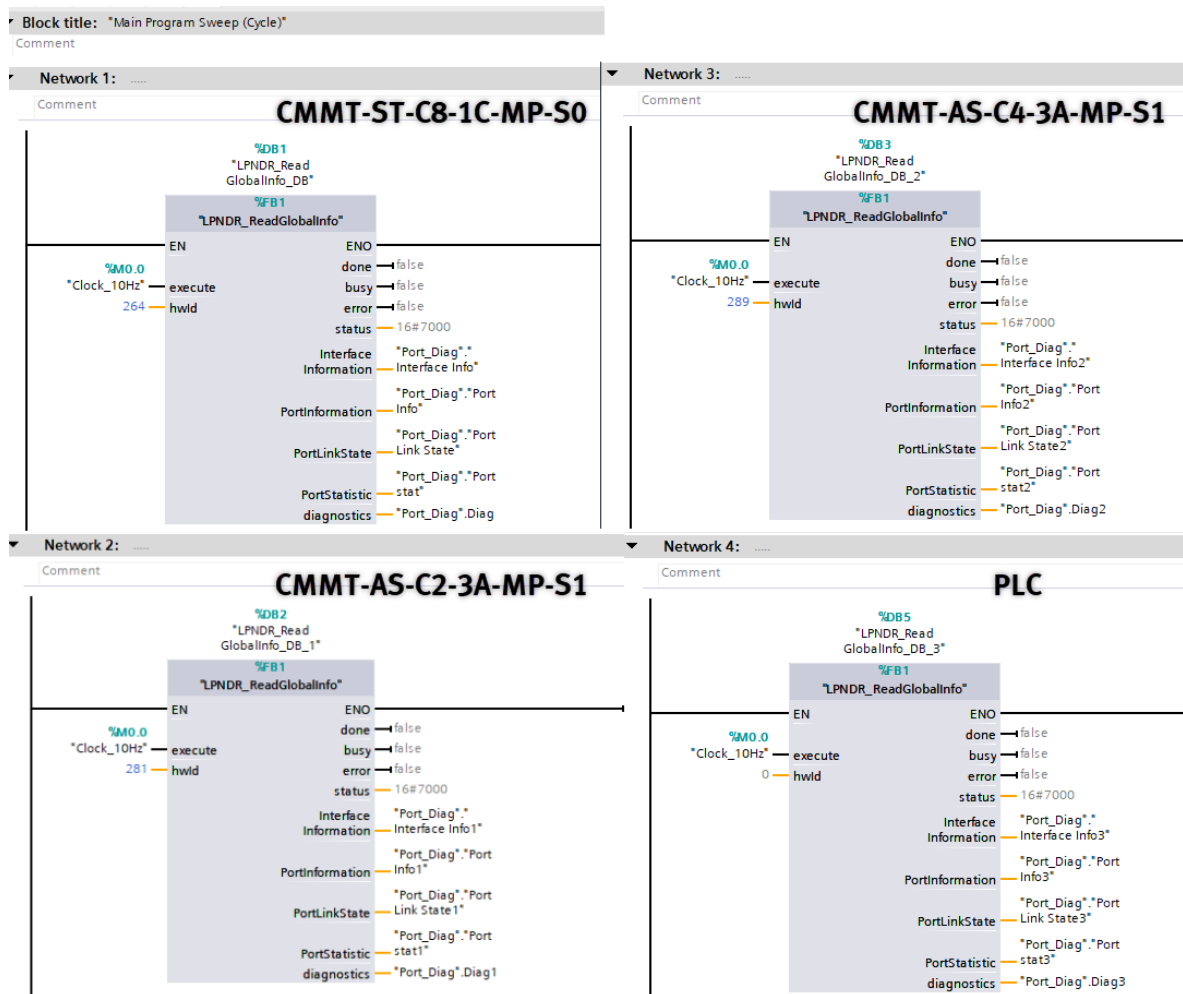


- For input tags, user can either enter them manually, or make tag so that they can be manipulated when PLC is online.

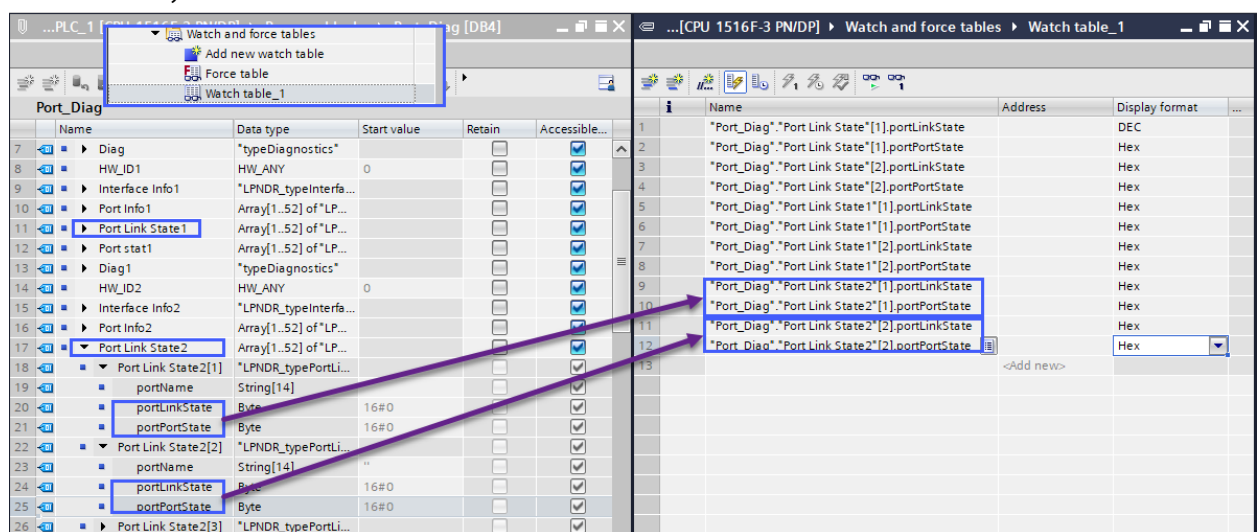


- For execute, user can make one tag to power the Function Block(User can assign clock pulse input for continuous trigger for the purpose of this application note).

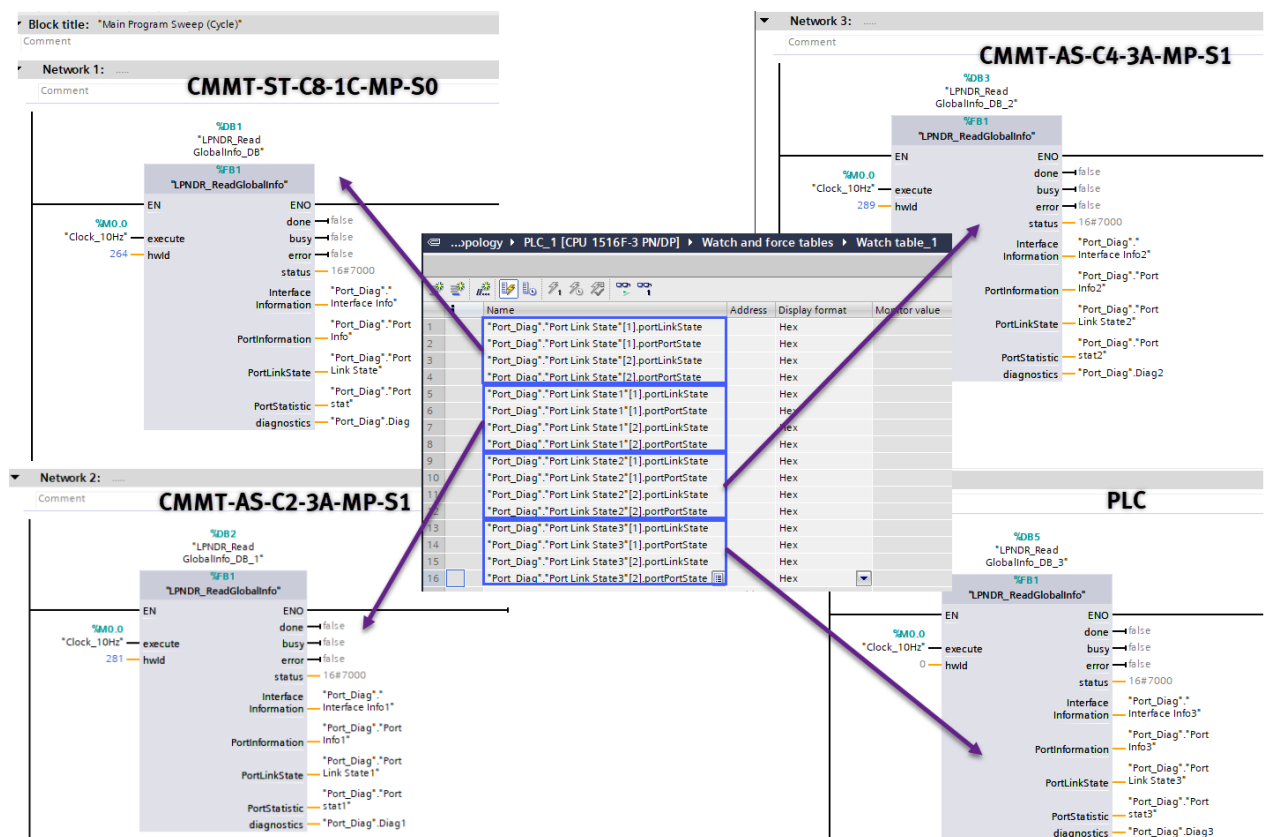
9. Functions blocks will look like image below for CMMT-ST and both the CMMT-ASs after all the configurations are done.



10. In order to monitor Ports, user can make a **Watch table**, and Drag and drop **portLinkState** and **portPortState** from Port Info from Data Block to watch table (Display format can be HEX or DEC as per the choice).



11. As shown in the image below, **Port Link state** is taken from Data block with respect to each module.



12. Now, Project can be compiled and Downloaded into PLC.



Note

- User can add more of Function blocks, as per the requirements, for Application note purpose, 4 sets of function blocks have been used to get Port diagnostics of 3 CMMT drives and PLC connected in the ring.
- PortLinkState [1]** corresponds to first port, here XF1. Whereas, **PortLinkState[2]** corresponds to XF2 for respective modules.

6.2 Sample code for Port Diagnostics using the Function Block

1. **Go Online**, then start monitoring **Main [OB1]** and **Watch table**.
2. In the image below, user can see that CMMT-ST and PLC connection (Especially from monitor value, connection from PLC side is disconnected) is not there, so, user can monitor link states through Port value in watch table as well as in **Topology view**.

The screenshot displays the configuration of the **LPNDR_ReadGlobalInfo** function block. The **EN** input is set to **TRUE**, and the **ENO** output is **TRUE**. The **done** output is **FALSE**, **busy** is **FALSE**, and **error** is **FALSE**. The **status** output is **16#0000**. The **Interface Information** output is **"Port_Diag".Interface Info**. The **PortInformation** output is **"Port_Diag".Port Info**. The **PortLinkState** output is **"Port_Diag".Port Link State**. The **PortStatistic** output is **"Port_Diag".Port stat**.

The Watch table for **CMMT_Ring_topology** shows the following data:

Name	Dis...	Monitor value	Modify value	Comment
"Port_Diag".Port Link State[1].portLinkState	Hex	16#02		
"Port_Diag".Port Link State[1].portPortState	Hex	16#00		
"Port_Diag".Port Link State[2].portLinkState	Hex	16#01		
"Port_Diag".Port Link State[2].portPortState	Hex	16#00		
"Port_Diag".Port Link State1[1].portLinkState	Hex	16#01		
"Port_Diag".Port Link State1[1].portPortState	Hex	16#00		
"Port_Diag".Port Link State1[2].portLinkState	Hex	16#01		
"Port_Diag".Port Link State1[2].portPortState	Hex	16#00		
"Port_Diag".Port Link State2[1].portLinkState	Hex	16#01		
"Port_Diag".Port Link State2[1].portPortState	Hex	16#00		
"Port_Diag".Port Link State2[2].portLinkState	Hex	16#01		
"Port_Diag".Port Link State2[2].portPortState	Hex	16#00		
"Port_Diag".Port Link State3[1].portLinkState	Hex	16#02		
"Port_Diag".Port Link State3[1].portPortState	Hex	16#00		
"Port_Diag".Port Link State3[2].portLinkState	Hex	16#01		
"Port_Diag".Port Link State3[2].portPortState	Hex	16#05		

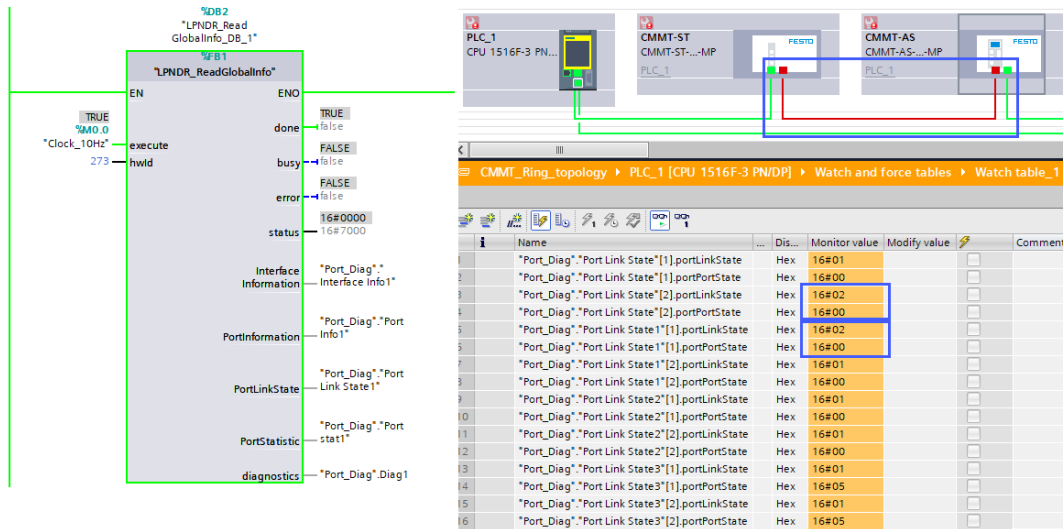
3. In the image below, the problem has been resolved with connection in topology view and Monitor values has gone back to normal.

The screenshot displays the resolved connection in the topology view. The connection is now established, and the monitor values are back to normal.

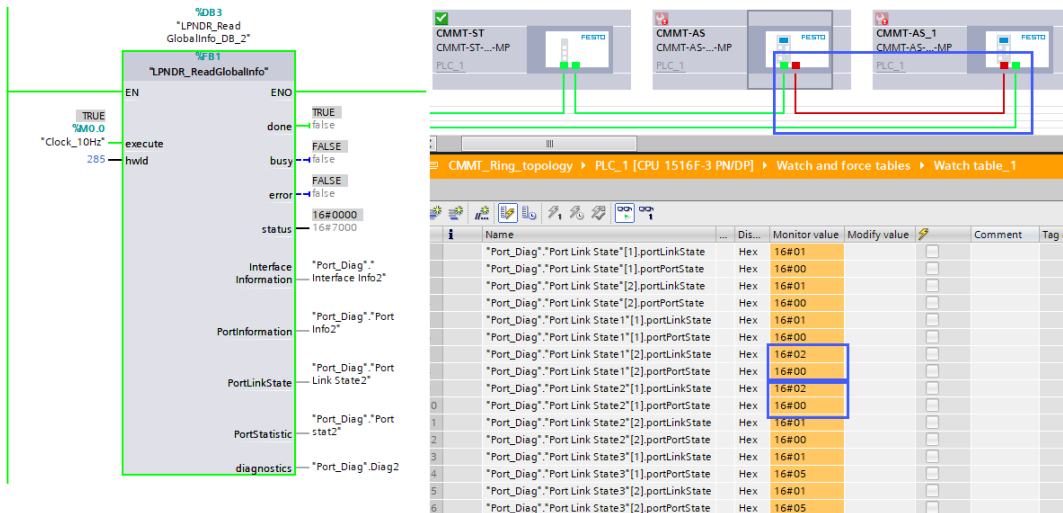
The Watch table for **CMMT_Ring_topology** shows the following data:

Name	Dis...	Monitor value	Modify value	Comment	Tag comment
"Port_Diag".Port Link State[1].portLinkState	Hex	16#01			
"Port_Diag".Port Link State[1].portPortState	Hex	16#00			
"Port_Diag".Port Link State[2].portLinkState	Hex	16#01			
"Port_Diag".Port Link State[2].portPortState	Hex	16#00			
"Port_Diag".Port Link State1[1].portLinkState	Hex	16#01			
"Port_Diag".Port Link State1[1].portPortState	Hex	16#00			
"Port_Diag".Port Link State1[2].portLinkState	Hex	16#01			
"Port_Diag".Port Link State1[2].portPortState	Hex	16#00			
"Port_Diag".Port Link State2[1].portLinkState	Hex	16#01			
"Port_Diag".Port Link State2[1].portPortState	Hex	16#00			
"Port_Diag".Port Link State2[2].portLinkState	Hex	16#01			
"Port_Diag".Port Link State2[2].portPortState	Hex	16#00			
"Port_Diag".Port Link State3[1].portLinkState	Hex	16#01			
"Port_Diag".Port Link State3[1].portPortState	Hex	16#02			
"Port_Diag".Port Link State3[2].portLinkState	Hex	16#01			
"Port_Diag".Port Link State3[2].portPortState	Hex	16#05			

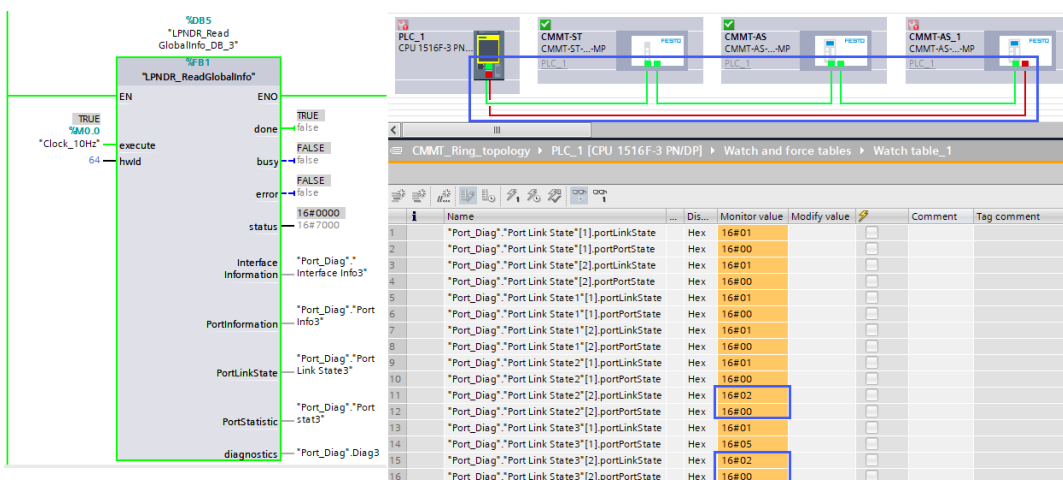
4. Trying with connection between CMMT-ST and CMMT-AS, feedback as shown in image below will be given.



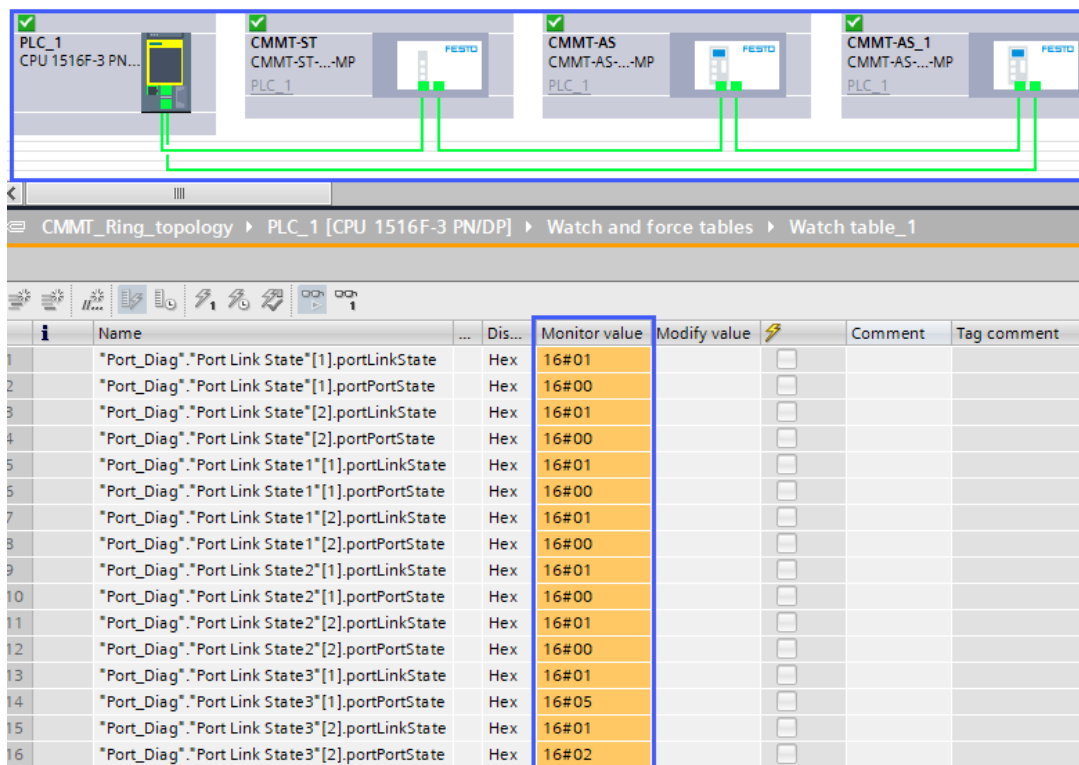
5. Similar with connection between both the CMMT-ASs.



6. Connection between second CMMT-AS_1 and PLC.



7. When there is no problem with any connection or the ports, it will be as shown in the image below.



8. Check the below tables (from Library for PROFINET data records (From siemens support)) in order to know what each value in monitor table represents.

Value of "portLinkState"	Link status
1	Link is up
2	Link is down
3	Port is in "testing" status
4	Link status is unknown
5	Port is in "demand" mode
6	Link status is not available
7	Port status is "LowerLayerDown"

Table 6.1

Value of "portPortState"	Link status
1	Port is in state "Disabled/ Discarding"
2	Port is in state "Blocking"
3	Port is in state „Listening“
4	Port is in state "Learning"
5	Port is in state "Forwarding"
6	Port is in state "Broken"
7	reserved

Table 6.2



Note

- One thing to be note on watch tables, **portPortStates** of CMMT stays 0, whereas PLC ports are able to give **Port State** values.